

Banking on Forest*

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Abstract

This paper examines how banks price firms' exposure to deforestation risk in syndicated loans. We distinguish between wildfire-driven forest loss (physical risk) and human-induced loss (transition risk). Using global loan-level data from 2002–2024 combined with geospatial forest loss measures, we find that banks charge higher spreads to forest-dependent firms following wildfire shocks. In contrast, human-induced loss is priced only after the introduction of the European Union Deforestation Regulation (EUDR), with stronger effects for EU banks and EU-exposed borrowers. Firms subsequently reduce reliance on forest-based inputs from high-risk suppliers, initiate reforestation, and divest from pollutive assets. Our findings highlight how banks transmit environmental risks and regulatory shocks to capital allocation and firm behaviour.

Keywords: deforestation, bank lending, environmental policy, green transition

JEL Classification: G21, G28, G30, Q54

1 Introduction

Forest loss can be the reason for economic loss in the form of natural disasters, or the consequence of economic activities such as agricultural land use and urbanization. The former, mostly driven by wildfires, accounts for nearly a quarter of global forest loss, represents an *acute physical risk* that immediately reduces the ecosystem services essential to a firm's production.^{1 2} In contrast, human-induced forest loss, mainly driven by the conversion of forests into agricultural land, contributes to *chronic physical risks* by driving long-term climate change, including increased carbon emissions and desertification (e.g., Van der Werf et al. 2009, Pan et al. 2011, Houghton et al. 2012).

Various international organizations have introduced initiatives and frameworks to encourage the green transition toward less forest-degrading production and supply chains.³ Among these, the European Union Deforestation Regulation (EUDR)⁴ mandates due diligence for commodities linked to deforestation.⁵ The regulation creates transition risk for firms whose production processes are tied to deforestation, through increased compliance costs, potential litigation, and disruptions in supply chains. These developments raise an important question: how do financial institutions respond to firms' exposure to deforestation-related risks?

Given firms' exposure to forest-loss related climate risks (FSB 2017), banks, as primary creditors, must not only ensure regulatory compliance (see, e.g., Allen et al. 2024, De Haas 2023) and address scrutiny from their own shareholders (see, e.g., Krueger et al. 2020, Giglio et al. 2021), but also actively manage and adjust their lending portfolios. This paper examines how banks incorporate firms' exposure to deforestation risk into syndicated loan contracts, distinguishing between physical risk arising from wildfire-driven forest loss and transition risk associated with human-induced deforestation.

¹ Studies show that over the past several decades, wildfire patterns have intensified regionally due to climate change, not past land uses (e.g., Westerling et al. 2006, Jolly et al. 2015).

² For instance, the California wildfires in 2018 caused an estimated \$148.5 billion damages (1.5% of state's GDP), including \$27.7 billion capital losses, \$32.2 billion health costs and \$88.6 billion indirect losses (Wang et al. 2021).

³ For example, REDD+ was launched under the 2013 UNFCCC Warsaw Framework, and the 2016 OECD-FAO Guidance aids agricultural risk management. Popp et al. (2014) estimate that, without land-use regulations, human-induced forest loss could contribute 13% of the global carbon budget needed to meet the Paris Agreement's 2°C target by 2050. With conservation measures like REDD+, this could be reduced to 7%.

⁴ EUDR's timeline: framework proposal (July 2019), legislative proposal (Nov 2021), agreement (Dec 2022), and entry into force (June 2023), with compliance due by Dec 2025 for large firms and June 2026 for small ones.

⁵ Under EUDR Article 2(15), an operator is any entity placing or exporting regulated products. This includes firms transforming one Annex I (regulated) product into another, such as cocoa butter into chocolate. https://green-business.ec.europa.eu/deforestation-regulation-implementation_en.

We compile a global sample from 2002 to 2024 from various datasets. The information on syndicated loan comes from DealScan. We use forest loss geospatial data from Global Land Analysis and Discovery (GLAD) laboratory at University of Maryland (Hansen et al. 2013; Tyukavina et al. 2022) and geocode firms' headquarters within a 10-kilometer radius to differentiate fire-induced and human-induced forest loss. We assess firms' reliance on forest-related ecosystem services with the tool of Exploring Natural Capital Opportunities, Risks and Exposure (ENCORE), which evaluates economic activities based on their dependency on ecosystem services and their impact on natural capital. Our final sample consists of 42,590 loan issuances, covering 6,329 borrowers and 1,298 lenders, with significant representation from the US, EU, and other OECD countries.

Forest loss arises from two distinct sources with fundamentally different implications for firms and their creditors. Wildfire-driven forest loss represents a realization of physical risk, as it can disrupt production and degrade the ecosystem services firms depend on. In contrast, human-induced forest loss exposes firms to transition risk, as it signals regulatory scrutiny, supply-chain disruption, and compliance costs. This distinction suggests that banks may differentiate between realized physical shocks and forward-looking regulatory risks when pricing corporate loans.

A key empirical challenge in measuring firms' exposure to forest loss is geographic alignment. While we proxy exposure using forest loss in the vicinity of firm headquarters, production activities may be spatially dispersed, introducing measurement error that is likely more pronounced in manufacturing and resource-intensive sectors. At the same time, these sectors are precisely those with the strongest economic reliance on forest-related inputs. In contrast, service-sector activities are less tied to physical locations, and environmental exposure operates primarily through indirect channels such as supply chains and regulatory expectations. As such, we expect the effects of forest loss on loan pricing to be concentrated in forest-dependent sectors, despite potential geographic measurement noise. The cross-sectional variation allows us to distinguish economically meaningful exposure from generic local shocks.

Our empirical results support this interpretation. First, banks charge higher loan spreads to forest-dependent firms following wildfire-induced forest loss, with an increase of 12 to 65 basis points depending on the severity of the event. This effect is consistent with the pricing of realized physical risk and reflects the impact of wildfire shocks on firms' short-term operations and liquidity. Second, human-induced forest loss does not affect loan pricing in the pre-EUDR period, but becomes economically and statistically significant following the introduction of the EUDR. A one-standard-deviation increase in human-induced loss is associated with a 7 basis

point increase in loan spreads for forest-dependent firms in the post-EUDR period. Year-by-year regressions further suggest that yield spreads for firms experiencing human-induced loss began rising following the 2016 Paris Agreement and became significantly positive from 2020 onward, aligning with the timeline of the EUDR proposal.

We further show that this transition-risk pricing is strongest for lending relationships involving EU banks and EU-exposed borrowers, and increases as the regulation moves from proposal to enforcement. These patterns suggest that regulatory developments play a central role in shaping how banks assess and incorporate environmental risks into lending decisions. In contrast, the pricing of wildfire-induced loss remains stable over time, consistent with its exogenous nature and limited connection to regulatory risk.

We next examine heterogeneity across banks and firms. Banks that publicly disclose commitments to deforestation-related issues exhibit stronger pricing responses to human-induced forest loss in the post-EUDR period. On the borrower side, firms that disclose deforestation-related policies face smaller increases in borrowing costs, suggesting that transparency may mitigate perceived transition risk.

To assess whether forest loss materially disrupts firm operations, we examine changes in cash flows following major fire-induced and human-induced forest loss events. We find that large fire-induced loss is associated with a decline in firm's operating cash flows, suggesting tangible operational impacts, while large human-induced loss has no significant effect on short-term cash flows, suggesting limited immediate impact on operations. These findings suggest that banks raise lending interest rates in response to fire-induced loss because it reflects real impact on short-term liquidity, whereas the absence of a similar response to human-induced loss is consistent with its lack of short-term operational consequences.

Finally, we provide supporting evidence on firms' responses to increased financing costs. Firms obtaining loans after human-induced forest loss shift their sourcing toward suppliers in countries with lower deforestation risk, engage in reforestation activities, and divest pollutive assets. In contrast, we find limited evidence of such adjustments following wildfire-induced loss. While these results are not the focus of our identification, they are consistent with the interpretation that loan pricing reflects broader transition pressures rather than purely contemporaneous operational shocks.

This paper contributes to several strands of literature. First, it introduces deforestation risk as a distinct dimension of climate-related financial risk and distinguishes between physical and transition channels in a unified empirical framework. By linking these environmental disruptions to widely recognized climate risks, we show that financial institutions and firms

acknowledge their threats and respond through risk management strategies and operational adjustments. This contributes to the literature on environmental action versus greenwashing (e.g., Kacperczyk and Peydró 2022, Giannetti et al. 2024, Sastry et al. 2024). In addition, we find forest-dependent firms' disclosures on addressing deforestation issues mitigate the loan pricing difference, aligning with Carbone et al. (2022), who show that firms' stating emission and reduction targets in disclosures is linked to lower credit risk, particularly for firms with more ambitious commitments.

Second, it provides evidence that banks incorporate environmental risks into loan pricing, extending the literature on climate risk and lending. A few prior studies suggest that banks price climate physical risks such as natural disasters. For instance, Brown et al. (2021) show that after severe winter weather, banks respond to firms' increased demand for capital by raising loan costs. Besides, Correa et al. (2022) find that loan spreads rise even for firms not directly impacted by natural disasters but with a high likelihood of exposure, suggesting that higher borrowing costs stem from lenders adjusting their expectations of climate risks. We show that forest-dependent firms affected by fire-induced forest loss face a 12-65 bps higher yield spread, economically more significant than differences due to relationship lending (10-17 bps gap between relationship and non-relationship lending in Bharath et al., 2011). In addition, our work is related to the literature of how banks mitigate physical risks in lending, including banks' divestment (e.g., Blikle et al. 2021, Ilabaca et al. 2024) and price adjustment (e.g., Javadi and Masum 2021, Nguyen et al. 2022, Götz et al. 2024) in syndicated loans and mortgages, to manage the credit risks from firms affected by natural disasters.

Third, this paper highlights the role of regulation in shaping capital market practices and leveraging financial institutions as key agents for policy implementation. This underscores how regulatory frameworks and environmental policies, such as the EUDR, influence risk assessment and capital allocation. This observation echoes the prior findings that climate initiatives and regulations spur lenders to re-allocate capital and tighten the loan terms. For example, Kacperczyk and Peydró (2022) show that banks re-allocate credit from brown to green firms after committing to carbon neutrality Science Based Targets initiative (SBTi); Degryse et al. (2023) document that banks offer cheaper loans to green firms following the Paris Agreement; Ivanov et al. (2024) find that cap-and-trade bills tighten bank loan terms for high-emission firms.

Finally, we provide direct firm-level evidence that climate regulations like the EUDR can drive operational changes, prompting firms to adopt more environmentally sustainable

practices. This demonstrates the tangible impact of financial and regulatory pressures in steering corporate behavior toward greener outcomes. This finding aligns with the existing literature showing that regulatory pressures push firms toward greener production and investment strategies. For instance, Goetz (2019) finds that firms reduce toxic emissions when lower capital costs enable greater investment in pollution prevention. Accetturo et al. (2022) show that increased credit supply raises the likelihood of firms' green investments in cleaner production technologies. Apicella and Fabiani (2023) document that firms exposed to carbon pricing expand credit demand and reduce emissions, highlighting how financing facilitates greener operations.

The remainder of the paper is structured as follows: Section 2 describes the data and sample. Section 3 outlines the empirical model and specifications, while Section 4 presents and discusses the empirical results. Section 5 concludes.

2 Data and sample construction

We compile a loan-level sample (2002-2024) from multiple sources for our main analyses. Geospatial data on global forest loss are obtained from the Global Land Analysis and Discovery (GLAD) laboratory at the University of Maryland (Hansen et al., 2013; Tyukavina et al., 2022). To capture changes in vegetation greenness, we use the NASA MODIS Normalized Difference Vegetation Index (NDVI). Information on ecosystem dependencies of production processes is sourced from the ENCORE (Exploring Natural Capital Opportunities, Risks and Exposure) tool, maintained and regularly updated by the United Nations. Customer–supplier relationships for U.S. firms are retrieved from Compustat Segment. Climate risk indices are obtained from Li et al. (2024). Corporate disclosures related to deforestation are collected via Refinitiv Advanced Filings Search (AdvFil). Data on plant divestitures are drawn from the EPA's Toxics Release Inventory (TRI) Program and the SDC M&A database. Syndicated loan information is sourced from DealScan, while firm and bank financials are compiled from Compustat Global, Compustat North America, and Refinitiv. We exclude financial firms, regulated utilities, and public administration firms.

We match the new LoanConnector DealScan data with the legacy DealScan dataset using the WRDS mapping utility. To obtain ISIN codes, the old DealScan data are linked to Worldscope via the DealScan–Worldscope Link Table (Beyhaghi et al., 2021). For borrower financial data, we use the link table provided by Schwert (2018) to merge DealScan with Compustat. Lender information is matched using the link table developed by Chava and Roberts (2008). To link Compustat company records with customer identifiers from Compustat

Segment, we rely on the mapping tables contributed by Cohen and Frazzini (2008) and Cen (2017).

The combined dataset, used for testing our main hypotheses, is structured at the loan level⁶, and contains 42,590 observations across 6,329 unique borrowers. Of these, 2,866 firms (45%) are headquartered in the US, 829 (13%) in the EU, and 4,766 (75%) in OECD member countries. This sample also includes 1,298 unique lenders, with 328 (25%) in the US, 226 (17%) in the EU, and 782 (60%) in OECD countries.

2.1 Measuring forest loss surrounding firm using geospatial data

We measure firm-level forest loss using two geospatial datasets from the University of Maryland. The Global Forest Change dataset (Hansen et al. 2013) provides annual 30-meter resolution data on gross forest cover loss from 2000 to 2023, with forest loss defined as a stand-replacement disturbance (from a forest to non-forest state). Tyukavina et al. (2022) disaggregate this into fire-induced and other causes. Fire-induced loss includes only full canopy fires, excluding mechanical burnings for agriculture. We define firm-level fire-induced and human-induced forest loss accordingly, with the latter primarily linked to commodity and agricultural drivers according to the classification of Curtis et al. (2018).⁷

Next, we measure the forest loss, both fire and human-induced, surrounding the headquarters of firms in our sample. Firm headquarters data (from Compustat and Refinitiv) are geocoded using a multi-step Google Maps algorithm to minimize location errors. We then calculate forest loss within a 10-kilometer radius of each headquarter via Google Earth Engine, recording fire- and human-induced loss in square kilometers.

Before merging with loan-level data, we match forest loss to the DealScan firm-year panel to analyze its distribution and create lagged measures. Figure 1 maps firm locations and 2023 forest loss intensity for the matched sample. Table 1 ranks average losses by country: Greece, Portugal, and Russia lead in fire-induced loss (U.S. ranks seventh), while Malaysia ranks highest in human-induced loss. This reflects Malaysia's globally leading deforestation rate from commercial agriculture and land use change.⁸ Panel A and Panel B of Table 2 show

⁶ We follow Chakraborty et al. (2018) to rank and select the lead arrangers with significant share in a deal. The final data for analysis is at the lead arranger-deal-earliest tranche (origination date) level.

⁷ Curtis et al. (2018) attribute about 77% of global forest loss to commodity and agricultural drivers, 23% to wildfires, and less than 1% to urbanization.

⁸ For instance: (1) Malaysia has the world's highest deforestation rate, as revealed by Google's forest map. <https://news.mongabay.com/2013/11/malaysia-has-the-worlds-highest-deforestation-rate-reveals-google-forest-map/>; (2) Causes of rainforest deforestation in Malaysia. <https://www.internetgeography.net/topics/causes-of-rainforest-deforestation-in-malaysia/>.

forest loss exposure by SIC division (excluding financials, utilities, and public administration). All industries are exposed to some degree of forest loss, though variation is greater for fire-induced loss; wholesale trade is most exposed to fire-induced loss, whereas mining firms are most exposed to human-induced loss.

[Figure 1] & [Table 1 and Table 2]

Table 3 presents summary statistics for fire-induced and human-induced forest loss lagged by one year in the DealScan loan-level sample. On average, firms experience 0.008 km² of fire-induced forest loss with a standard deviation of 0.123, and 0.170 km² of human-induced loss with a standard deviation of 0.419.

[Table 3]

2.2 Measuring firms' production forest-dependency

We use ENCORE (Exploring Natural Capital Opportunities, Risks and Exposure), a tool developed in 2018 by Global Canopy, UNEP FI, and UNEP-WCMC, to assess firms' reliance on forest-based ecosystem services. ENCORE includes two pathways: dependency (direct reliance) and impact (pressure on ecosystems). We use dependency to capture forest risk. In our sample, 80% of above-median dependency firms also show above-median impacts, underscoring the dual role firms play as both reliant on and contributors to forest risk.

ENCORE rates 271 production activities across 25 ecosystem services using a six-level materiality scale (from very low to very high), based on literature reviews and expert validation.⁹ For each activity, it provides: (1) a materiality rating on the production process dependency on various ecosystem services¹⁰, (2) the natural capital assets required for each ecosystem service and the associated benefits, and (3) the natural or human-induced pressures (e.g., forest loss drivers) affecting these assets' ability to provide goods and services.

In our analysis, we begin by identifying ecosystem services provided by forest-related natural capital assets (referred to as "*forest services*" afterwards for simplicity). For instance, the ecosystem service "climate regulation" depends on the proper functioning of habitats, soils, water, and wetlands. These natural capital assets are classified as forest-related because they are affected by deforestation drivers such as habitat modification and overharvesting. In this case, "climate regulation" will be identified as *forest services*. After selecting only *forest*

⁹ Where no links are identified, the activity-service pair is marked as N/A (Not Applicable) or ND (No Data).

¹⁰ For example, the production of large-scale rainfed arable crops depends on 18 ecosystem services. It is highly dependent on animal-based energy due to low resilience to disruption, but shows minimal reliance on water quality, which does not critically affect operations.

services, we then aggregate the six-level materiality ratings (we re-assign values from 0-5) of the production process dependency on the *forest services* into the production process level. This approach specifically focuses on highly forest-related ecosystem services to assess production process-level risks, rather than broadly aggregating all ecosystem service dependency ratings.

Finally, we aggregate the production process-level dependency scores into the GICS industry level, which is the original classification used by ENCORE. We then use the industry crosswalk tables provided by ENCORE, and the ISIC-USSIC link table, to match the data with our sample at the 2-digit SIC level. After excluding financial firms, regulated utilities, and public administration, we ended up with 64 unique industry groups. We normalize the continuous dependency score into 0-5 scale, to be comparable to the ENCORE six-level materiality rating at the production process level.

Table 2 Panel C shows the summary statistics of the dependency score for each high-level SIC division at the firm level. The top three high-level SIC divisions with the highest average forest dependency scores are agriculture, forestry, and fishing (2.52), mining (2.31), and manufacturing (1.04). At the 2-digit SIC level, top five industries with the highest dependency score are forestry (5.0), food and kindred products (3.15), coal mining (2.79), agricultural production crops (2.71), and agriculture production livestock and animal specialties (2.71). Table 3 shows that the average dependency measure *Dependency* is 0.91 for the loan-level sample.

3 Empirical design

We begin by examining whether banks price in physical and transition risks from forest loss in loans. Firms more reliant on forest-based inputs face greater risk when nearby loss occurs, with the risk type depending on its cause. Specifically, fire-induced forest loss represents a realization of *physical risk*, as it can abruptly disrupt firms that rely on forests by unexpectedly removing raw materials essential to their production processes. Wildfires may also produce secondary effects, such as smoke damage or destruction of physical assets, that can impact a broader range of firms. However, these effects are typically shorter-lived and less directly tied to production activities. In contrast, human-induced forest loss may similarly reduce the availability of forest-based inputs, but it often stems from the firm's own actions, such as supply chain expansion or land-use change. Consequently, it exposes forest-dependent firms to *transition risks*, as the loss signals involvement in deforestation and environmental degradation. This increases the likelihood of regulatory scrutiny under current or anticipated

deforestation-related policies. Thus, we estimate the following model specification at the loan level:

$$\begin{aligned} \text{Yield spread}_{b,f,l,t} = & \beta_1 \text{Loss}_{f,t-1} \times \text{Dependency}_i + \beta_2 \text{Loss}_{f,t-1} + \beta_3 \text{Dependency}_i \\ & + \theta_1 (\text{Loan ctrls})_{b,f,l,t} + \theta_2 (\text{Bank ctrls})_{b,t-1} + \theta_3 (\text{Firm ctrls})_{f,t-1} \quad (1) \\ & + FE + \epsilon_{b,f,l,t}. \end{aligned}$$

The dependent variable *Yield spread*_{b,f,l,t} is “all-in-spread drawn” (AISD) divided by 100. *Loss*_{f,t-1} captures the total forest loss, both fire-induced and human-induced, within a 10-kilometer radius of a firm’s headquarter in the previous year.¹¹ *Dependency*_i is the proxy for the levels of an industry’s forest dependency, aggregated at the 2-digit SIC level. Alternatively, we also construct a country-weighted forest dependency measure, where industry dependency is weighted by country-year-level total forest loss, to account for heterogeneity in deforestation exposure across countries.¹² The coefficient β_1 captures the differential change in loan yield spreads for high forest-dependent firms relative to low forest-dependent firms in response to forest loss, enabling us to assess whether banks price risk based on firms’ reliance on forest-related inputs.

The vector *FE* represents a set of fixed effects to account for unobserved heterogeneity. In our baseline specification, we include year fixed effects to control for time-varying macroeconomic conditions, and high-level SIC division fixed effects to control for persistent sector-level heterogeneity in business models, exposure to deforestation risk, and loan pricing.¹³ In other specifications, we further include industry $\times$ year fixed effects to absorb time-varying shocks within broad sectors. To account for cross-border lending heterogeneity, we

¹¹ While using headquarters locations may introduce measurement error when production occurs elsewhere, prior studies employing this proxy argue that a substantial portion of firms’ operations, business activities and employees are concentrated near their headquarters (e.g., Chaney et al. 2012, Korniotis and Kumar 2013, Barrot and Sauvagnat 2016, Tuzel and Zhang 2017, Huynh and Xia 2021). At the time of drafting, we are unable to conduct plant-level robustness check using U.S. plant-level data from the Census of Manufactures or Annual Survey of Manufactures, as access is suspended under federal data restrictions. To further mitigate this concern, we also examine two cleaner subsamples in later robustness analyses: borrowers matched to Orbis manufacturing firms classified as headquarters with no branches, and relatively small firms in the bottom decile of total assets.

¹² We calculate the country-weighted dependency score as $\text{Dependency} \times (1 + \text{forest loss}_{c,t})$, where both components are normalized between 0 and 1. This ensures that in years with minimal forest loss, the original industry dependency remains unchanged, while in years with high country-level forest loss, the weighted score can rise to a maximum of 2, effectively doubling the base dependency to reflect elevated risk.

¹³ Although average dependency is highest in *agriculture, forestry, and fishing*, dependency is not concentrated exclusively at the high-level sector level. As shown in Table 2 Panel C, the average dependency score in this division is higher than in mining by 0.167 (SE = 0.0168) and higher than in manufacturing by 1.418 (SE = 0.0298). At the same time, there is substantial within-industry variation: in *agriculture, forestry, and fishing*, dependency ranges from 1.622 to 5, with a standard deviation of 0.932, which exceeds the mean dependency score in construction, the fourth-highest SIC division.

also include bank $\times$ firm country fixed effects in another specification, which absorbs persistent bilateral differences in lending practices arising from cross-country variation in regulatory regimes, deforestation exposure, and credit market conditions. Loan-level controls include a set of dummy variables: *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower-level controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*, all lagged by one year. Lender-level controls include *Bank size*, also lagged by one year. The summary statistics for the variables are reported in Table 3. All variables are defined in Appendix Table A.1. $\epsilon_{b,f,l,t}$ is the error terms, clustered at the 2-digit SIC industry and year level to account for correlated shocks within industries over time.

Next, we examine whether banks increase lending rates in response to the rising transition risks from deforestation, using the introduction of the EUDR as a policy shock that amplifies such risks. A key regulatory distinction is the source of forest loss: natural events like wildfires are beyond firm control and unlikely to trigger penalties, while human-induced loss, such as from business expansion or land-use change, represents a local environmental exploitation and is more likely to attract exposure to regulatory scrutiny, especially for forest-dependent firms. Unlike voluntary initiatives, the EUDR imposes legally binding due diligence, requiring firms to ensure their products are deforestation-free. It applies not only to EU firms but also to those trading with the EU, with strict traceability rules linking products to their geographic origin.

To capture rising transition risk, we define the period following July 2019, when the EUDR legislative framework was proposed, as the beginning of heightened regulatory pressure on deforestation. This timing reflects the forward-looking nature of market responses to anticipated compliance obligations. Accordingly, we estimate the following loan-level model:

$$\begin{aligned}
\text{Yield spread}_{b,f,l,t} = & \beta_1 \text{Loss}_{f,t-1} \times \text{Dependency}_i \times \text{Post EUDR}_t \\
& + \beta_2 \text{Loss}_{f,t-1} \times \text{Dependency}_i + \beta_3 \text{Loss}_{f,t-1} \times \text{Post EUDR}_t \\
& + \beta_4 \text{Dependency}_i \times \text{Post EUDR}_t \\
& + \beta_5 \text{Loss}_{f,t-1} + \beta_6 \text{Dependency}_i + \beta_7 \text{Post EUDR}_t \\
& + \theta_1 (\text{Loan ctrls})_{b,f,l,t} + \theta_2 (\text{Bank ctrls})_{b,t-1} + \theta_3 (\text{Firm ctrls})_{f,t-1} \\
& + \text{FE} + \epsilon_{b,f,l,t}.
\end{aligned} \tag{2}$$

Post EUDR_t is a time indicator equal to one after July 2019, and zero otherwise. A significantly positive β_1 would suggest that banks adjust loan pricing in response to heightened transition risk from human-induced loss under stricter regulatory scrutiny.

4 Empirical results and discussions

4.1 Baseline results: forest loss and loan spreads

We estimate Equation (1) to test whether banks increase loan spreads for forest-dependent firms following nearby forest loss. Table 4 presents the results for both measures of *Fire loss* and *Anthropogenic loss*. Column (1) shows that at the sample mean of *Dependency* (0.91), one additional square kilometer of fire loss corresponds to a 14-basis-point increase in spreads. In contrast, column (2) shows the interaction of *Dependency* and *Anthropogenic loss* is small and not significant. Column (3) confirms the observations with both loss types included. These results suggest that banks distinguish between the two forest loss types and primarily price unanticipated physical risks from fire-induced loss.¹⁴

[Table 4]

A potential concern is that the estimated effect may reflect time-varying shocks to a small set of high-dependency industries rather than differential pricing of deforestation risk within industries. To address this concern, column (4) replaces industry fixed effects with industry×year fixed effects, absorbing all common shocks at the SIC-division-year level. The coefficient on the interaction term remains economically and statistically similar, increasing slightly from 0.415 to 0.463. This pattern indicates that the estimated pricing effect is not driven by contemporaneous shocks common to high-dependency industries.

Next, we ask whether banks condition loan pricing on borrowers' country-level deforestation risk. To address this, column (5) employs a country-weighted dependency measure, assigning greater weight to firms in high-risk countries. The interaction coefficient rises from 0.415 to 0.554, indicating banks are more responsive to physical risk in vulnerable geographies. Column (6), which includes bank×firm country fixed effects, yields a similar coefficient (0.596), suggesting pricing is driven by within-country variation rather than cross-country differences. That is, banks respond to localized fire-induced loss rather than uniformly penalizing all firms in fire-prone countries. Together, these results imply banks incorporate both systemic country-level risk and idiosyncratic local shocks in loan pricing.

¹⁴ We do not find corresponding effects on non-price loan terms. In both the baseline specification and the post-EUDR specification, neither loan amount nor maturity differs significantly across firms with different levels of dependency following either loss measure. This pattern suggests that the intensive margin adjustment operates primarily through pricing rather than through other contract terms.

Given that the forest loss measure is constructed around firm headquarters, we conduct robustness tests in subsamples where headquarters location is more likely to align closely with the borrower's actual operations. Table B.1 replicates the baseline specification in Equation (1). Column (1) restricts the sample to borrowers matched to Moody's Orbis manufacturing firms classified as single headquarters with no branches, while column (2) restricts the sample to small firms, defined as those in the bottom decile of total assets. In both specifications, the interaction between *Dependency* and *Fire loss* remains positive and statistically significant. At the sample mean of *Dependency* (0.91), a one-standard-deviation increase in fire-induced loss corresponds to a 39-basis-point increase in loan spreads in the single-HQ sample and a 19-basis-point increase in the small-firm sample.

Another potential concern is that loan pricing reflects general climate risk rather than forest-specific exposure. Table B.2 addresses this by controlling for firm-level acute, chronic, and transition risks (Li et al. 2024) in a sample of U.S. listed firms. The *Dependency* and *Fire loss* interaction remains stable (0.653 to 0.651) and significant at the 1% level, indicating the effect is not confounded by broader climate risk but reflects the unique pricing of forest-related physical shocks.

4.2 Robustness to alternative forest loss measures

The severity of fire-induced forest loss likely shapes banks' risk assessments. While small fires may be inconsequential, extreme events can disrupt operations significantly for forest-dependent firms. If banks perceive such tail risks as disproportionately costly, loan pricing should exhibit nonlinearity. For robustness, Table B.3 re-estimates Equation (1) by interacting *High dependency* (firms above the median forest dependency) with discrete fire loss indicators defined by percentile thresholds, with *Yield spread* as the dependent variable.

Column (1) shows that a one-standard-deviation increase in fire-induced loss raises spreads by 2.3 bps for high-dependency firms (interaction = 0.190). This likely understates tail risk due to right-skewed fire loss distribution. Column (2) uses a binary indicator for any fire loss; the interaction coefficient is 0.146, implying a 14.6 bps spread difference when any fire loss occurs. Columns (3)-(6) apply increasingly severe percentile cutoffs, with interaction effects strengthening as fire magnitude intensifies. The top panel of Figure 2 further confirms a nonlinear pricing pattern: interaction coefficients between *High dependency* and fire loss indicators rise steadily across percentile thresholds (50th to 99th), with a sharp increase beyond the 95th of fire loss. Across specifications, coefficients range from 0.12 to 0.65, implying that

forest-dependent firms face 12-65 bps higher spreads following fire events, depending on severity.

[Figure 2]

4.3 Policy shock: the introduction of the EUDR

To assess whether banks adjust pricing in response to deforestation-related transition risk, we estimate Equation (2) using the global sample first, focusing on the post-EUDR period (August 2019 to December 2024). Column (1) of Table 5 shows that the *Fire loss* and *Dependency* interaction remains positive and significant, while the triple interaction with *Post EUDR* is not, suggesting no change in pricing of fire-induced loss after the regulation. Column (2), using *Weighted dependency*, yields similar non-significance, consistent with fire-induced loss being exogenous and thus not triggering transition risk concerns.

[Table 5]

In comparison, columns (3) and (4) of Table 5 examine *Anthropogenic loss*. The triple interaction with *Dependency* and *Post EUDR* is 0.223 in column (3), implying a 7.0 bps spread increase for a one-standard-deviation rise in both dependency and human-induced loss. Using *Weighted dependency* in column (4), the coefficient rises to 0.356, suggesting banks impose larger penalties on forest-dependent firms in high-risk countries post-EUDR. Figure B.1 presents year-by-year estimates of Equation (1) using rolling five-year windows. The top panel shows that the pricing effect of fire-induced loss is generally positive, with a small dip and increasing variation after 2014. In contrast, the bottom panel shows that pricing effects for human-induced loss decline after 2008, rise steadily post-2016 around the Paris Agreement, and become significantly positive from 2020 onward, coinciding with the EUDR proposal timeline.

To mitigate concerns that the headquarter-based forest loss measure may not fully capture firms' main operations, Table B.1 applies the same sample restrictions to the EUDR tests using the single-HQ sample in column (3) and the small-firm sample in column (4). The triple interaction term remains positive and statistically significant in both specifications. Economically, a one-standard-deviation increase in both *Dependency* and *Anthropogenic loss* is associated with a 47-basis-point increase in loan spreads in column (3) and a 101-basis-point increase in column (4).

4.3.1 Country heterogeneity

To further assess whether EUDR-driven pricing effects are stronger within the EU, Table 6 compares EU lender-borrower pairs with other lending situations. Column (1), which

focuses on EU bank-EU firm loan subsample, shows a triple interaction coefficient of 1.529 between *Anthropogenic loss*, *Dependency* and *Post EUDR*, indicating that EU banks impose higher spreads on forest-dependent EU firms exposed to human-induced loss post-EUDR. This effect is significantly larger than the 0.198 coefficient for non-EU pairs (column 2). In economic terms, the post-EUDR pricing gap between high- and low-dependency firms (relative to the sample median) ranges from approximately 40 to 200 bps depending on the level of human-induced loss, as shown in the bottom panel of Figure 2.

[Table 6]

We also test for spillovers to non-EU OECD countries. Column (3) shows a triple interaction of 0.384 for OECD lending pairs that exclude any EU lender or borrower, roughly one-fourth the EU effect in column (1). This is significantly larger than the 0.047 coefficient for non-OECD pairs (column 4). While weaker than within-EU pricing, the results indicate OECD lenders are responsive to EU deforestation policy signals.

To isolate the compliance role of EU banks, we then focus on EU operators: firms either headquartered in the EU or with major EU customers (identified via Compustat Segment). This ensures direct exposure to EUDR. Table B.4 re-tests on this subsample and shows that EU banks significantly raise spreads for forest-dependent firms facing anthropogenic loss post-EUDR, while non-EU banks show no response.

Given the multiple phases of EUDR implementation, we next examine whether changes in regulatory certainty affect loan pricing. From the framework proposal in July 2019 to enforcement in June 2023, firms faced elevated but uncertain transition risk. Once the regulation took effect, compliance terms became clearer, but pressure intensified as deadlines approached. Table B.5 uses two post indicators and presents how banks adjust loan spreads across these phases. We define post indicator *Post EUDR (phase 1)* as August 2019 to June 2023, and *Post EUDR (phase 2)* as the period following enforcement. Column (1) of Table B.5 shows that for EU bank-firm pairs, the triple interaction of *Anthropogenic loss*, *Dependency* and *Post EUDR* rises from 1.972 in phase 1 to 4.000 in phase 2. These correspond to yield spread increases of 38.0 and 40.7 bps, respectively, for a one-standard-deviation rise in both dependency and human-induced loss during phases 1 and 2. These results suggest that the differential pricing strengthens as the regulation moves from proposal to enforcement. Column (2) finds no meaningful response for loans involving non-EU parties. Columns (3) and (4), using *Weighted dependency*, show even larger effects for EU pairs in both phases, further supporting the stronger compliance role of EU banks.

4.3.2 Credit supply side: compliance role of committed banks

From credit supply side, a natural question is whether banks that publicly commit to addressing deforestation play a more active role in pricing transition risk. Table 7 presents the results by re-estimating the interaction between *Anthropogenic loss* and *Dependency* across four subsamples: committed and non-committed banks, before and after the EUDR. Committed banks are identified as those who have mentioned “deforestation” in their public disclosures, including ESG reports, annual reports, and SEC filings retrieved from Refinitiv AdvFil.

[Table 7]

In the post-EUDR period, only committed banks exhibit a significant positive interaction coefficient of 0.150, implying that these banks charge higher spreads to forest-dependent firms following human-induced forest loss. Economically, the estimate implies that a 1 km² increase in loss is associated with a 23-basis-point increase in loan spreads for a one-unit increase in *Dependency*. No significant pricing response is observed for non-committed banks or in the pre-EUDR period. These findings suggest that committed banks internalize regulatory transition risks more actively in loan pricing, highlighting a compliance-oriented channel on the credit supply side.

4.3.3 Firm heterogeneity: firm commitments on deforestation

We next examine whether this pricing effect is mitigated when firms actively engage with deforestation issues in their public disclosures. Similarly, Table 8 presents the results by re-estimating the *Anthropogenic loss* and *Dependency* interaction across firm commitment status and pre- versus post-EUDR subsamples. Committed firms are defined as those who mention “deforestation” in public filings (via Refinitiv AdvFil again) during the previous year, capturing time-series variation in firm-level deforestation exposure and engagement. The proxy rests on two assumptions. First, “deforestation” is a specific term rarely used outside environmental contexts, unlike broader terms such as “ecosystem”. Second, firms referencing “deforestation” in formal disclosures typically signal recognition or mitigation efforts, rather than admitting involvement.

[Table 8]

Columns (1) and (2) report results for committed firms before and after the EUDR, and columns (3) and (4) show results for non-committed firms. Only non-committed firms in the post-EUDR period exhibit a significant positive interaction between forest dependency and human-induced loss, implying that lenders impose higher borrowing costs on uncommitted firms when deforestation risk is salient. The estimates imply that in the post-EUDR period, a

1 km² increase in loss raises loan spreads by 30 basis points only for non-committed firms for each one-unit increase in *Dependency*. The absence of such pricing for committed firms suggests that disclosure serves as a mitigating signal of transition risk.¹⁵

5 Mechanisms and ex-post outcomes

5.1 Mechanism: forest loss and firm operation

To assess whether loan pricing reflects realized operational disruption, we estimate dynamic difference-in-differences regressions where the outcome is operating cash flow divided by tangible assets of the prior year. Firms are classified using *Top dependency*, equal to one for the top 30 percent of forest reliance and zero for the bottom 30 percent. *Post large fire loss* and *Post large anthropogenic loss* are event-time indicators spanning from two years before to four years after the loss. Large loss events are defined as those in the top 20 percent of the annual distribution, measured consistently across all distance bands from 10 to 80 kilometers around the firm, to ensure the event is both nearby and severe. We exclude cases with overlapping events to avoid contamination. Figure 3 plots the dynamic interaction between *Top dependency* and each forest loss type. Following large fire-induced loss, high-dependency firms experience a sharp decline in operating cash flow that drops below minus 20 percent relative to low-dependency peers. The negative effect persists for two years before gradually recovering by year four. In contrast, human-induced loss generates no immediate impact. Instead, cash flows among high-dependency firms begin to rise over time, with relative performance gains becoming evident by the fourth post-event years, suggesting business expansion-driven gains.

[Figure 3]

5.2 Ex-post outcome in production

After receiving syndicated loans, firms may adjust their sourcing or reforestation activities, offering insight into whether lending supports green transitions or sustains deforestation-intensive practices. If high-risk firms reduce deforestation exposure post-loan, this suggests lender engagement in promoting sustainability. Conversely, continued deforestation implies that lending may enable environmentally harmful operations. We begin by examining changes in forest-related supply.

¹⁵ As a robustness check, we replicate the analysis using firms' E pillar scores from Refinitiv ESG rating as an alternative measure of environmental engagement and obtain similar results.

Table 9 reports firm-level results on how receiving loans after forest loss affects a firm's supply chain. We use two outcome variables to capture different types of supply chain transitions: (1) *Supply dependency* measures the share of a firm's inputs sourced from forest-dependent suppliers, calculated as the sum of each supplier's *Dependency* weighted by its share of the firm's total purchases. A decline in this measure indicates a shift away from forest-based inputs, reflecting an overall transition in production sourcing. (2) *Country-adj supply* measures the share of inputs from forest-dependent suppliers located in high-deforestation-risk countries, calculated as the sum of each supplier's *Weighted dependency* (accounting for both forest dependency and country-level deforestation risk) multiplied by its share of the firm's total purchases. A decrease in this variable reflects a shift toward responsible sourcing, where the firm continues to use forest inputs but increasingly sources them from countries with lower deforestation risk. In both Panel A and Panel B, columns (1) and (3) report average outcomes over three years forward, while columns (2) and (4) report four-year averages. Firms are classified using *If get loan*, a dummy equal to one if a firm obtains a syndicated loan in the same year or within one year following a large forest loss, and zero otherwise.

[Table 9]

Panel A examines firm responses to large human-induced forest loss. *Post large anthropogenic loss* equals one for the three years after the event and zero for the three years before. In columns (1) and (2), the interaction between *If get loan* and the time indicator is negative and weakly significant, suggesting limited reduction in overall reliance on forest-based inputs. Columns (3) and (4) show stronger effects: the interaction terms are negative and significant, indicating that firms shift toward sourcing from lower-risk countries rather than reducing total forest input use. This implies that loans are associated with more responsible sourcing under heightened transition risk. Panel B evaluates responses to large fire-induced loss using the same specification. *Post large fire loss* is defined over the same three-year window. Across all columns, the interaction terms are negative but not significant, suggesting no material adjustment in supply chains among firms receiving loans. This lack of response reflects the more exogenous nature of wildfire shocks, which do not intensify transition risks in the same way as human-driven deforestation.

5.3 Ex-post outcome in reforestation

We next assess whether firms receiving syndicated loans after large human-induced forest loss engage in reforestation. Unlike supply chain shifts, reforestation requires active restoration, as the environmental damage is long-lasting and cannot be reversed through

sourcing changes alone. To measure reforestation, we use the *Normalized Difference Vegetation Index* (NDVI), a satellite-based indicator of vegetation greenness from NASA MODIS. Table 10 reports results from a firm-year panel. The time indicator *Post large anthropogenic loss* equals one for the three years after the event and zero for the three years before. *If get loan* equals one if a firm receives a syndicated loan in the same year or within one year of the loss event.

[Table 10]

Panel A of Table 10 reports the results for the full sample. The interaction between *If get loan* and *Post large anthropogenic loss* is positive and significant across all windows, with coefficients from 1.145 (one-year forward outcome) to 1.184 (two-year forward outcome). This indicates greater reforestation following loan receipt. Panel B restricts the sample to firms above the median in forest dependency. Coefficients are ranging from 1.724 to 1.677, implying that forest-dependent firms respond more through reforestation. To test whether this response is specific to human-driven loss, Table B.6 repeats the analysis for fire-induced events. No significant relationship is found between loan receipt and NDVI, consistent with wildfires being external shocks. This supports the interpretation that reforestation is more likely when deforestation is anthropogenic and linked to transition risk.

5.4 Ex-post outcome in pollutive plant divestiture

Firms may also respond to deforestation-related transition risk by adjusting their asset portfolios. One channel is divesting pollutive plants, particularly if syndicated loans provide the capital to support such restructuring. If loans facilitate this transition, we would expect a higher likelihood of divestiture. Table 11 tests this by examining whether firms receiving loans are more likely to divest pollutive forest-linked plants after large human-induced forest loss. The sample includes U.S. public firms with at least one TRI-listed plant. Divestiture data come from the SDC M&A database. We limit the sample to transactions where the buyer acquires over 50 percent ownership and exclude financial firms as acquirers or targets. Pollutive plants are restricted to only forest-dependent ones. The divestiture indicator is multiplied by 100, so coefficients represent percentage-point changes in divestiture likelihood. The outcome is measured forward from one to four years across columns.

[Table 11]

The results show that firms receiving loans after human-induced forest loss are more likely to divest pollutive forest-dependent plants. Starting from the forward two-year outcome (column 2), the triple interaction of *Anthropogenic loss*, *Dependency*, and *If get loan* is positive

and significant, with coefficients rising from 1.107 to 1.234 as the outcome window extends from $t+2$ to 4. For instance, column (3) implies that a one-standard-deviation increase in both forest dependency and loss raises divestiture probability by 0.35 percentage points for loan-receiving firms relative to others.

5.5 Further discussions: selection into loan receipt and alternative measures

A potential concern is that firms receiving loans differ systematically from non-recipients, potentially biasing post-loan outcomes. To assess this, the t-test in Table B.7 shows that loan receipt is associated with higher book debt, confirming expanded overall debt capacity. Table B.8 estimates a probit model of loan receipt. In column (3), the coefficients of *Anthropogenic loss* and *Dependency* are positive, with *Anthropogenic loss* weakly significant at the 10% level, suggesting limited evidence that deforestation exposure may affect selection into syndicated borrowing. To further address selection, Table B.9 includes the Inverse Mills Ratio in the analysis of production outcomes, with no change in results. Similar robustness holds for reforestation (Table B.10) and divestiture outcomes (Table B.11).

Another concern is that NDVI may not directly capture firm-level reforestation, especially if projects occur off-site, such as through carbon offsets. To address this, we construct an alternative outcome based on self-reported reforestation. Using a keyword dictionary derived from voluntary “Forestry and Land Use” carbon offset classifications, we flag firm disclosures from Refinitiv sources when they mention reforestation.¹⁶ The variable *Reforest* equals 100 if such activity is reported within the post-loan estimation window.

Table B.12 tests this alternative outcome and confirms prior findings. Forest-dependent firms receiving loans after human-induced loss are more likely to report reforestation, with the triple interaction of *Anthropogenic loss*, *Dependency*, and *If get loan* positive and significant across all models. Coefficients range from 0.549 to 0.734. Column (2) implies a one-standard-deviation increase in both variables raises reforestation disclosure likelihood by 0.23 percentage points. Results remain robust to selection controls, as shown in Table B.13.

¹⁶ The dictionary is based on the categories of voluntary “Forestry & Land Use” projects documented in the Voluntary Registry Offsets Database (Haya et al. 2025). The dictionary includes terms related to key project types: afforestation, reforestation, (improved) forest management, (avoided) forest conversion, forest restoration, forest projects, (improved) grassland management, (avoided) grassland conversion, grassland restoration, (improved) wetland management, (avoided) wetland conversion, wetland restoration, and REDD+.

6 Conclusion

This paper examines how banks price syndicated loans for firms exposed to deforestation risks, distinguishing between fire-induced and human-induced forest loss. We find that banks increase loan spreads for forest-dependent firms affected by fire-induced forest loss, charging a premium of 12-65 basis points. Regulatory frameworks, such as EUDR, have a profound influence on financial institutions' risk assessment and capital allocation. EU banks impose higher loan spreads on forest-dependent firms affected by human-induced loss compared to non-EU banks, with a stronger effect on firms operating within high-deforestation-risk countries. The regulatory impact extends to OECD countries, though with a lower intensity. Over different phases of EUDR implementation, loan pricing became more sensitive to transition risks from the policy framework proposal in July 2019 until the regulation's entry into force in June 2023, with a further rise afterwards.

We further find that firms also respond to the rising costs of capital and evolving regulations. Those that actively disclose deforestation-related risks experience reduced loan pricing penalties, demonstrating the role of transparency in mitigating financial burdens. In response to transition risks associated with human-induced forest loss, firms shift sourcing toward countries with lower deforestation risk, engage in reforestation efforts, and divest pollutive forest-dependent plants.

This study contributes to the literature by uncovering how banks integrate climate risks into loan pricing, highlighting the role of regulations in influencing capital markets, and providing firm-level evidence of regulatory-driven operational shifts.

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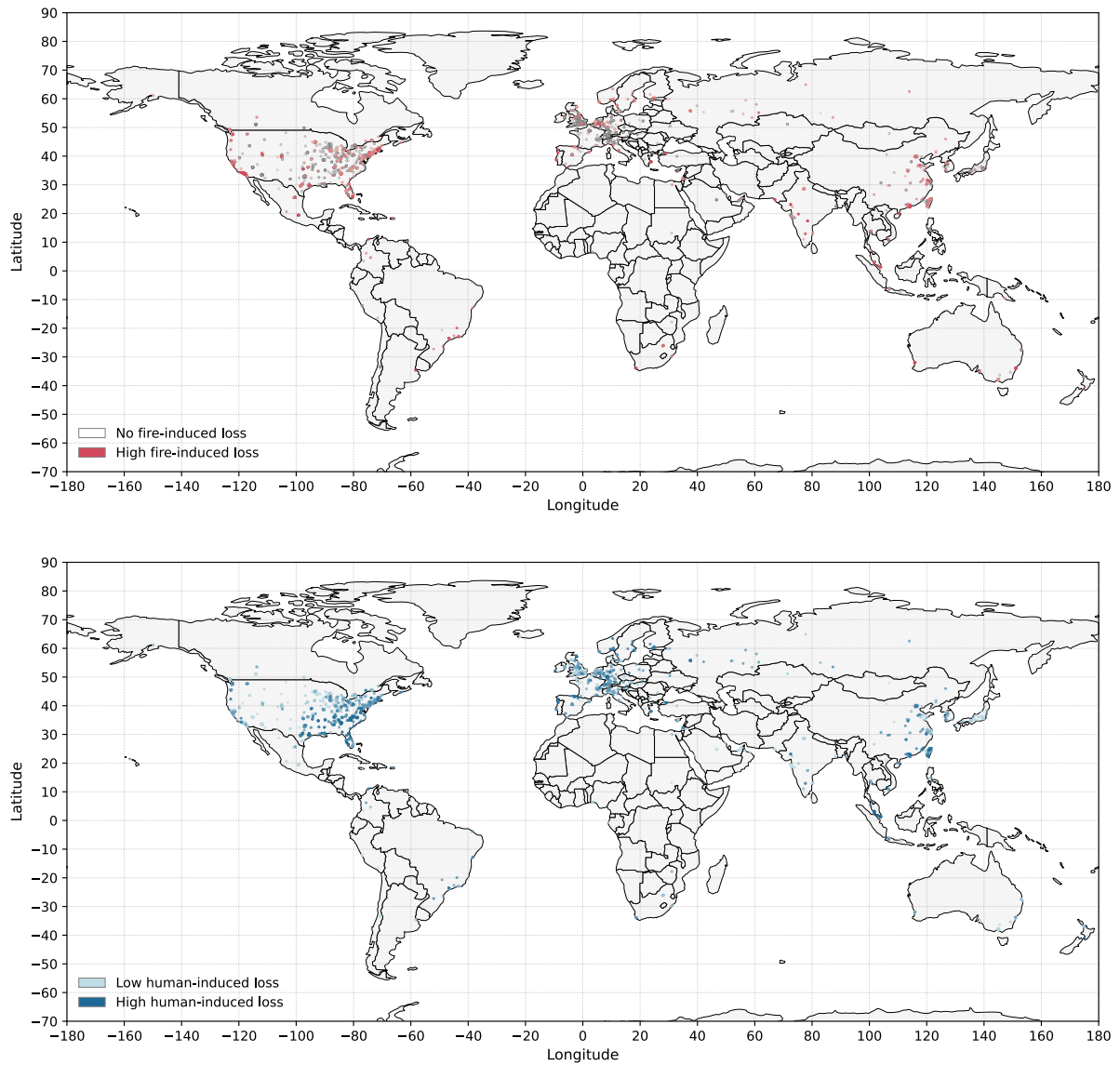


Figure 1. Forest loss of the sample firms in 2023

The plots illustrate the geographic distribution of our sample firms and the forest loss around them in 2023. The first figure depicts fire-induced forest loss, where gray circles indicate firms with no fire-induced loss, and increasing shades of red represent greater fire-induced loss. The second figure focuses on human-induced forest loss, with light blue circles indicating minimal forest loss and darker shades indicating more severe loss. The forest loss is measured within a 10km radius around each firm. The circles on the maps are visually enlarged to clearly display firms' locations, and do not represent the actual geographical areas of the forest loss.

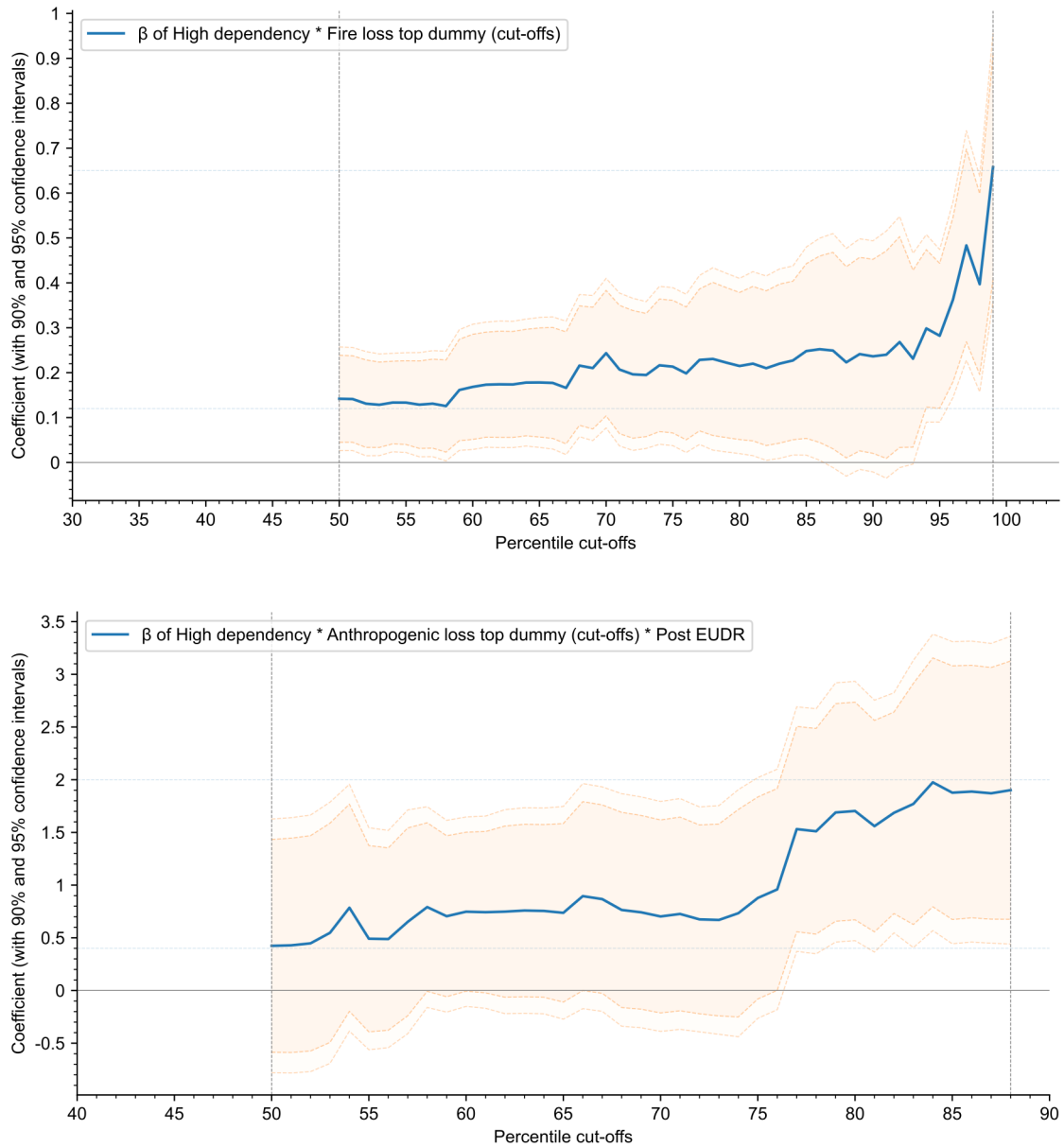


Figure 2. Forest loss thresholds and differential pricing effects

This figure plots estimated coefficients from interaction terms involving *High Dependency* and deforestation loss percentile cut-off dummies, with *Yield spread* as the dependent variable. The top panel shows the interaction with fire-induced loss; the bottom panel plots the triple interaction with human-induced loss and *Post EUDR*, using EU bank-firm lending pairs. *High Dependency* equals one if the firm's forest dependency score is above the median. Loss dummies equal one if exposure exceeds a given percentile threshold. Blue lines show coefficient paths; shaded areas indicate statistical significance (light orange for 5%, dark orange for 10%).

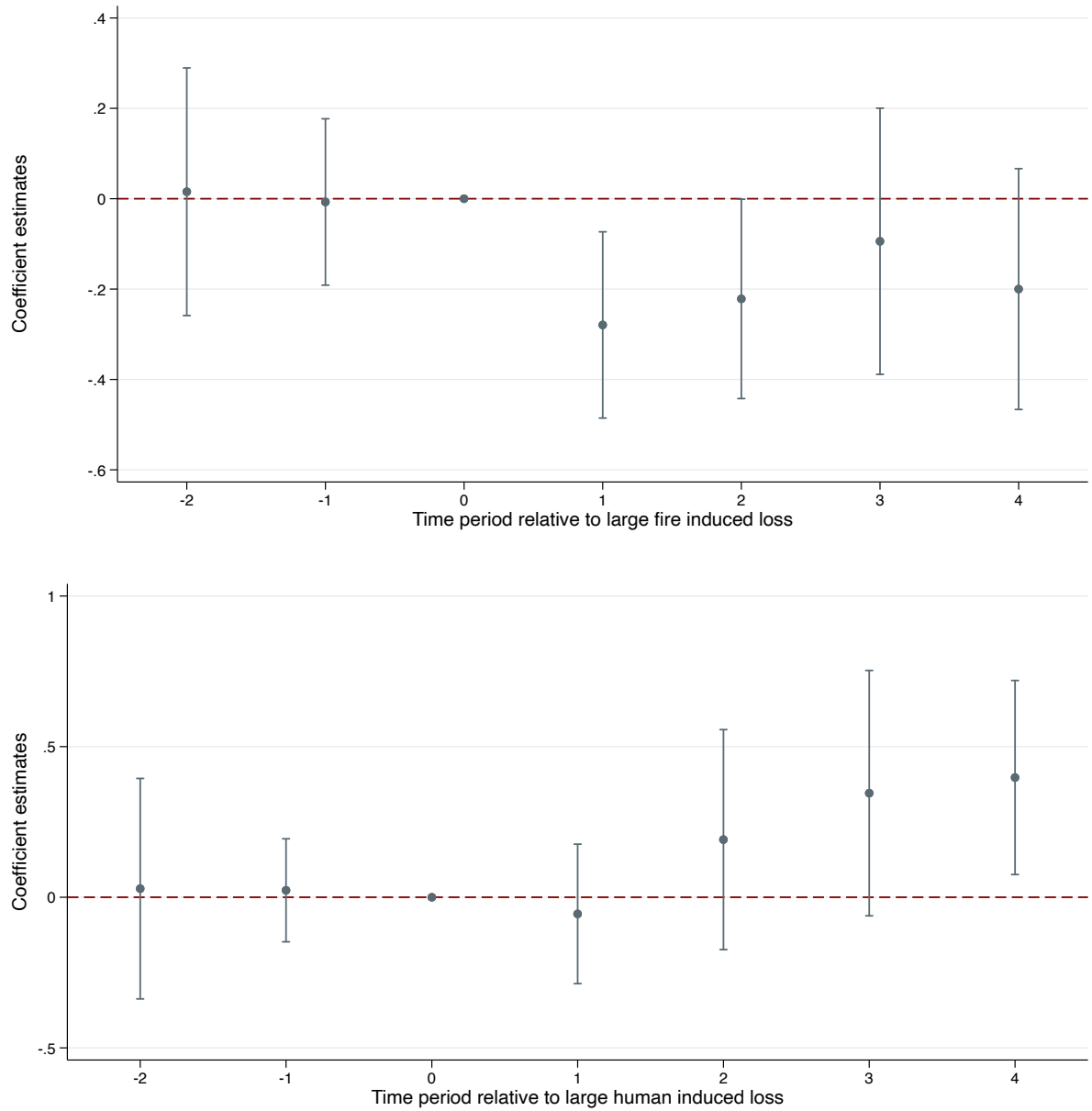


Figure 3. Dynamic effects of large forest loss on firm cash flow

This figure plots the time trend of the treatment effect estimates of firms' high forest-dependency on cash flow around large forest loss events. The top figure shows fire-induced forest loss, while the bottom figure shows human-induced forest loss. Cash flow is measured by dividing operating cash flow by the previous year tangible assets. Firms' dependency on forest is measured by *Top dependency*, a dummy variable equals one if the firm's forest-dependency score is within top 30% of the sample, and zero if within bottom 30%. For each period, we plot the point estimate (the solid circle) and the 90% confidence interval (the vertical lines intersecting the solid circles). Time indicators are defined for each year around a large forest loss event. For instance, Time (0) denotes the year of the large forest loss; Time (-1) denotes the year before the large forest loss, and so forth.

Table 1. Forest loss rankings across countries

This table reports the cross-country statistics of fire-induced forest loss (*Fire loss*) and human-induced forest loss (*Anthropogenic loss*) in square kilometers (km²), in the firm-year panel sample. Panel A lists the top ten countries with the highest average fire-induced forest loss. Panel B lists the top ten countries with the highest average human-induced forest loss. The selection condition requires that each country has more than 100 firms in the sample.

Panel A. Top ten countries (>100 firms) with the highest average forest loss from fire

<i>Fire loss</i>	Obs.	Mean	STD	Min	Median	Max
Greece	428	0.3249	1.4402	0	0.00283	24.2641
Portugal	165	0.1275	0.5805	0	0.00631	5.1818
Russia	811	0.07622	1.1142	0	0.00713	31.4377
South Africa	343	0.06176	0.3540	0	0.00545	5.7976
Australia	2,682	0.04276	0.0868	0	0.01562	1.7574
Colombia	151	0.04008	0.09676	0	0.00779	0.7654
United States	40,255	0.02335	0.7545	0	0.00075	96.6292
Italy	882	0.01755	0.2124	0	0	4.1066
Brazil	803	0.01571	0.08534	0	0.00248	2.0665
Mexico	895	0.01562	0.02622	0	0.00422	0.1647

Panel B. Top ten countries (>100 firms) with the highest average human-induced forest loss

<i>Anthropogenic loss</i>	Obs.	Mean	STD	Min	Median	Max
Malaysia	502	1.3599	1.3103	0.03571	0.9703	11.4485
Portugal	165	0.7898	1.1001	0.0049	0.2370	5.2686
Finland	540	0.5442	0.6275	0.01819	0.3229	4.1991
Spain	1,369	0.4701	1.1811	0	0.08622	9.9872
Norway	670	0.4691	0.5331	0	0.3549	3.6835
Singapore	999	0.4597	0.5784	0.00626	0.2681	5.3275
Sweden	1,002	0.3886	0.7078	0	0.2161	6.2322
Poland	128	0.3885	0.5769	0	0.1543	3.1312
Turkey	370	0.3588	0.5463	0	0.1488	4.3528
Vietnam	157	0.3437	1.1123	0	0.02197	7.0750

Table 2. Industry rankings for forest loss and dependency

This table reports the descriptive statistics and the high-level SIC division rankings for fire-induced forest loss (*Fire loss*), human-induced forest loss (*Anthropogenic loss*), and forest dependency score (*Dependency*) in the firm-year level panel sample. Panel A provides details on *Fire loss*, Panel B focuses on *Anthropogenic loss*, and Panel C on *Dependency*.

Panel A. Industry rankings of the forest loss from fire

<i>Fire loss</i>	Obs.	Mean	STD	Min	Median	Max
Wholesale Trade	4,430	0.05306	2.0547	0	0.00067	96.6292
Construction	3,385	0.03250	0.4782	0	0.00034	20.1115
Services	18,031	0.01665	0.2528	0	0.0007	16.8422
Retail Trade	6,485	0.01534	0.2957	0	0.00071	16.8789
Agriculture, Forestry, Fishing	560	0.01428	0.0940	0	0.00071	1.5693
Transp. & Comm., Electric, Gas	12,675	0.01392	0.2484	0	0.00055	24.2641
Mining	8,631	0.01341	0.0974	0	0.00073	3.6712
Manufacturing	46,464	0.01067	0.2572	0	0.00059	31.4377

Panel B. Industry rankings of human-induced forest loss

<i>Anthropogenic loss</i>	Obs.	Mean	STD	Min	Median	Max
Mining	8,631	0.2151	0.6158	0	0.04281	7.0488
Construction	3,385	0.2057	0.5824	0	0.0433	9.1428
Retail Trade	6,485	0.1984	0.4884	0	0.04472	7.7134
Wholesale Trade	4,430	0.1980	0.4576	0	0.04742	6.8372
Agriculture, Forestry, Fishing	560	0.1974	0.5211	0	0.04088	4.6086
Manufacturing	46,464	0.191	0.4959	0	0.04687	18.2719
Services	18,031	0.1788	0.4794	0	0.04317	20.0675
Transp. & Comm., Electric, Gas	12,675	0.1673	0.4404	0	0.03463	6.8476

Panel C. Firm-level summary of industry rankings of the dependency score

<i>Dependency</i>	# of firms	Mean	STD	Min	Median	Max
Agriculture, Forestry, Fishing	34	2.5188	0.9322	1.6228	2.693	5
Mining	560	2.3089	0.3264	1.7654	2.4649	2.7719
Manufacturing	2,730	1.039	0.7062	0.2281	0.8421	3.1316
Construction	203	0.8503	0.0726	0.5614	0.8684	0.8684
Services	1,265	0.5636	0.3199	0.114	0.7735	1.8041
Retail Trade	430	0.5548	0.1823	0.3596	0.5044	0.9561
Transp. & Comm., Electric, Gas	785	0.4999	0.5584	0	0.152	1.5298
Wholesale Trade	274	0.3395	0.0022	0.3377	0.3377	0.3421

Table 3. Summary statistics

This table reports the summary statistics of our main analysis sample at the loan level. The definition of variables is detailed in Appendix Table A.1.

Variable	Obs.	Mean	STD	Min	Median	Max
All in spread drawn bps	42,590	191.666	146.5982	15	150	825
Yield spread	42,590	1.9167	1.466	0.15	1.5	8.25
Fire loss (km ²)	42,590	0.0077	0.1228	0	0.0007	11.8811
Anthropogenic loss (km ²)	42,590	0.1696	0.4197	0	0.0461	10.2686
Dependency	42,590	0.9108	0.7472	0	0.7735	5
Weighted dependency	42,590	0.5852	0.5268	0	0.4517	3.8872
Loan amount (m\$)	42,590	645.9761	1,021.1591	2.5	250	5,000
Log loan amount	42,590	5.4331	1.5737	0.9163	5.5215	8.5172
Maturity (month)	42,590	54.6548	25.7748	3	60	222
Log maturity	42,590	3.8587	0.6031	1.0986	4.0943	5.4027
If secured loan	42,590	0.3731	0.4836	0	0	1
If base prime	42,590	0.0069	0.0827	0	0	1
If refinance	42,590	0.1719	0.3773	0	0	1
Repeated lending	42,590	0.395	0.4889	0	0	1
Loan purpose	42,590	0.2276	0.4193	0	0	1
Bank total assets (m\$)	42,590	111,081.01	219,229.38	0.4038	3,578.467	2,608,333.8
Bank size	42,590	9.1003	2.976	0.3392	8.183	14.7742
Total assets (m\$)	42,590	114,271.17	363,745.55	0.246	6,889.5908	1,846,191.1
Firm size	42,590	8.9617	2.3658	0.2199	8.8379	14.4286
Leverage	42,590	0.6131	0.2286	0.0276	0.6046	3.0809
ROA	42,590	0.0459	0.1119	-3.2236	0.0451	0.252
Liquidity	42,590	0.0886	0.088	0	0.0618	0.8015
Credit rating	42,590	5.6283	18.7277	0	0	98
Log credit rating	42,590	0.6205	1.2075	0	0	4.5951

Table 4. Forest loss and loan spreads

This table reports the baseline results of loan-level tests examining how forest loss affects yield spreads differently for firms based on their forest dependency. The dependent variable is *Yield spread*, measured by AISD divided by 100. *Fire loss* refers to firm-level forest loss from fire in the previous year, and *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. Columns (1) to (4) use *Dependency* to measure forest dependency at the 2-digit SIC level, and columns (5) to (6) use *Weighted dependency*, which adjusts for risks associated with country-year-level forest loss and is rescaled to a range of 0 to 5. Columns (1), (2), (3), (5), (6) include year fixed effects and industry fixed effects at the high-level SIC division level. Column (6) further includes bank×firm country-level fixed effects. Column (4) includes industry×year fixed effects. Definitions for all variables are provided in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC level and year level, with values reported in parentheses. Statistical significance is denoted by ***, **, and * at the 1%, 5%, and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>						
<i>Dependency measures</i>	<i>Dependency</i>			<i>Weighted dependency</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
Fire loss×	0.412**		0.415**	0.463**	0.554**	0.596**
<i>Dependency measure</i>	(0.170)		(0.177)	(0.170)	(0.218)	(0.264)
Fire loss	-0.238*		-0.246*	-0.270**	-0.269*	-0.332*
	(0.124)		(0.129)	(0.126)	(0.136)	(0.164)
Anthropogenic loss×		-0.008	-0.012	-0.006	-0.058*	-0.028
<i>Dependency measure</i>		(0.027)	(0.028)	(0.027)	(0.031)	(0.028)
Anthropogenic loss		0.055	0.057	0.039	0.085**	0.039
		(0.036)	(0.036)	(0.027)	(0.040)	(0.027)
<i>Dependency measure</i>	-0.002	0.002	0.000	-0.010	0.119*	-0.056
	(0.045)	(0.048)	(0.048)	(0.046)	(0.069)	(0.046)
If secured loan	0.667***	0.665***	0.665***	0.671***	0.662***	0.672***
	(0.071)	(0.071)	(0.071)	(0.072)	(0.071)	(0.055)
If base prime	1.016***	1.013***	1.013***	1.052***	1.002***	1.268***
	(0.225)	(0.227)	(0.226)	(0.193)	(0.223)	(0.280)
If refinance	-0.133*	-0.132*	-0.132*	-0.133*	-0.130*	-0.089
	(0.074)	(0.074)	(0.075)	(0.070)	(0.074)	(0.055)
Repeated lending	-0.246***	-0.247***	-0.247***	-0.237***	-0.246***	-0.088***
	(0.045)	(0.045)	(0.045)	(0.046)	(0.045)	(0.024)
Log loan amount	-0.146***	-0.146***	-0.145***	-0.145***	-0.143***	-0.099***
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.014)
Log maturity	0.934***	0.932***	0.933***	0.923***	0.933***	0.712***
	(0.154)	(0.154)	(0.155)	(0.148)	(0.153)	(0.105)
Loan purpose	-1.057***	-1.060***	-1.060***	-1.070***	-1.053***	-1.205***
	(0.213)	(0.213)	(0.213)	(0.219)	(0.212)	(0.221)
Firm size	-0.485***	-0.479***	-0.478***	-0.452***	-0.454***	-0.410*
	(0.152)	(0.151)	(0.151)	(0.148)	(0.156)	(0.236)
Leverage	0.042**	0.042**	0.041**	0.040**	0.039**	0.012
	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.017)
ROA	0.008	0.008	0.008	0.007	0.006	-0.005
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.013)
Liquidity	-0.076**	-0.077**	-0.076**	-0.078**	-0.078**	-0.168***
	(0.032)	(0.032)	(0.032)	(0.032)	(0.032)	(0.015)
Log credit rating	0.129***	0.129***	0.129***	0.126***	0.133***	0.127***
	(0.045)	(0.045)	(0.045)	(0.043)	(0.045)	(0.033)
Bank size	0.664***	0.664***	0.665***	0.691***	0.668***	0.529***
	(0.072)	(0.072)	(0.072)	(0.071)	(0.072)	(0.049)
Constant	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	No	Yes	Yes
Industry FE	Yes	Yes	Yes	No	Yes	Yes
Industry × year FE	No	No	No	Yes	No	No
Bank × firm country FE	No	No	No	No	No	Yes
Observations	42,590	42,590	42,590	42,590	42,590	42,590
Adjusted R-squared	0.357	0.357	0.357	0.375	0.358	0.518

Table 5. Transition risks from EU deforestation regulation change

This table presents loan-level regression results examining how the development of EUDR affects yield spreads for forest-dependent firms versus other firms when forest loss occurs. The dependent variable is *Yield spread*. Columns (1) and (2) use *Fire loss* as the loss measure, which is the firm-level forest loss from fire in the previous year. Columns (3) and (4) use *Anthropogenic loss*, which is the firm-level forest loss from human activities in the previous year. All columns use *Dependency* to measure forest dependency at the 2-digit SIC level. *Post EUDR* is a time indicator, defined as one after the deforestation framework stage (July 2019), and zero otherwise. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Loss measure</i>	<i>Fire loss</i>		<i>Anthropogenic loss</i>	
	(1)	(2)	(3)	(4)
Loss measure×Dependency× <i>Post EUDR</i>	-0.913 (2.441)		0.223* (0.120)	
Loss measure×Weighted dependency× <i>Post EUDR</i>		-1.915 (3.485)		0.356* (0.173)
Loss measure×Dependency	0.449** (0.204)		-0.025 (0.039)	
Loss measure×Weighted dependency		0.575** (0.244)		-0.072* (0.036)
Loss measure× <i>Post EUDR</i>	0.517 (1.913)	0.891 (2.010)	-0.358** (0.149)	-0.366** (0.147)
Dependency× <i>Post EUDR</i>	0.194** (0.089)		0.159 (0.097)	
Weighted dependency× <i>Post EUDR</i>		0.255* (0.134)		0.192 (0.155)
Loss measure	-0.257* (0.141)	-0.268* (0.144)	0.077* (0.043)	0.105** (0.044)
<i>Post EUDR</i>	-0.015 (0.107)	0.017 (0.109)	0.058 (0.136)	0.099 (0.138)
Dependency	-0.022 (0.048)		-0.014 (0.051)	
Weighted dependency		0.088 (0.065)		0.106 (0.067)
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	42,590	42,590	42,590	42,590
Adjusted R-squared	0.357	0.358	0.358	0.358

Table 6. Country heterogeneity of EUDR's effect on yield spreads

This table reports the loan-level results examining the country heterogeneity of the effect of the introduction of EUDR on yield spread differently for forest-dependent firms and other firms when human-induced loss occurred. The dependent variable is *Yield spread*. All columns use *Dependency* to measure forest dependency at the 2-digit SIC level, and use *Anthropogenic loss*, which is the firm-level forest loss from human activities in the previous year. *Post EUDR* is a time indicator, defined as one after the deforestation framework stage (July 2019), and zero otherwise. Columns (1) and (2) compare pairs of EU banks with EU firms versus other lending situations. Columns (3) and (4) compare pairs of OECD (excluding EU) banks and firms versus other situations. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Bank country – Firm country pair</i>	<i>EU</i>	<i>Non-EU</i>	<i>OECD</i>	<i>Non-OECD</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency× <i>Post EUDR</i>	1.529** (0.658)	0.198 (0.127)	0.384*** (0.130)	0.047 (0.218)
Anthropogenic loss×Dependency	-0.032 (0.167)	-0.032 (0.030)	-0.019 (0.022)	-0.036 (0.149)
Anthropogenic loss× <i>Post EUDR</i>	-1.445* (0.789)	-0.298* (0.159)	-0.560** (0.243)	0.046 (0.351)
Dependency× <i>Post EUDR</i>	0.076 (0.121)	0.155 (0.099)	0.077 (0.110)	0.200 (0.183)
Anthropogenic loss	0.125 (0.184)	0.063* (0.033)	0.042* (0.023)	0.003 (0.188)
<i>Post EUDR</i>	0.470*** (0.148)	0.043 (0.134)	0.340*** (0.101)	-0.313*** (0.091)
Dependency	-0.239*** (0.075)	0.023 (0.047)	0.011 (0.037)	-0.019 (0.115)
<i>Chi-square test</i>	(1) = (2)	6.787***	(3) = (4)	7.725***
<i>P-value</i>		0.009		0.005
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	5,518	37,072	17,464	12,418
Adjusted R-squared	0.488	0.357	0.423	0.240

Table 7. Loan pricing and bank commitment to deforestation

This table presents loan-level regression results testing whether banks' public commitments to deforestation influence the pricing of forest risk. The dependent variable is *Yield spread*. The sample is split by bank commitment status and regulatory period. *Committed banks* are defined as those that have mentioned "deforestation" in public disclosures, such as ESG reports, annual reports, and SEC filings retrieved from Refinitiv AdvFil. Columns (1) and (2) present the results for committed bank subsample, and columns (3) and (4) show the non-committed bank subsample. The pre-EUDR period includes loans from January 2014 to July 2019, and the post-EUDR period includes loans from August 2019 to December 2024. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Bank subsample</i>	<i>Committed banks</i>		<i>Non-committed banks</i>	
<i>Subsample period</i>	<i>Pre-EUDR</i>	<i>Post-EUDR</i>	<i>Pre-EUDR</i>	<i>Post-EUDR</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency	-0.078 (0.093)	0.150** (0.071)	-0.056 (0.061)	0.163 (0.207)
Anthropogenic loss	0.160 (0.114)	-0.226 (0.172)	0.179 (0.119)	-0.194 (0.291)
Dependency	-0.081 (0.078)	0.080 (0.057)	0.066 (0.065)	0.111 (0.094)
<i>Chi-square test</i>	(1) = (2)	2.594	(3) = (4)	0.750
<i>P-value</i>		0.107		0.387
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	3,094	2,932	6,626	2,385
Adjusted R-squared	0.397	0.456	0.302	0.419

Table 8. Firm commitments on deforestation

This table presents loan-level regression results examining how firm commitments on deforestation affect yield spreads for forest-dependent firms versus other firms when human-induced forest loss occurs. The dependent variable is *Yield spread*. *Committed firms* are defined as those that mention “deforestation” in the previous year in public disclosures, such as ESG reports, annual reports, and SEC filings retrieved from Refinitiv AdvFil. Columns (1) and (2) present the results for committed firm subsample, and columns (3) and (4) show the non-committed firm subsample. The pre-EUDR period includes loans from January 2014 to July 2019, and the post-EUDR period includes loans from August 2019 to December 2024. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Firm subsample</i>	<i>Committed firms</i>		<i>Non-committed firms</i>	
<i>Subsample period</i>	<i>Pre-EUDR</i>	<i>Post-EUDR</i>	<i>Pre-EUDR</i>	<i>Post-EUDR</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency	0.321 (0.670)	-0.374 (0.900)	-0.044 (0.065)	0.195** (0.091)
Anthropogenic loss	-0.141 (0.390)	0.372 (0.743)	0.152 (0.101)	-0.281 (0.187)
Dependency	0.283* (0.141)	0.257* (0.149)	0.017 (0.066)	0.109 (0.071)
<i>Chi-square test</i>	(1) = (2)	0.034	(3) = (4)	3.552*
<i>P-value</i>		0.853		0.060
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	270	291	9,450	5,026
Adjusted R-squared	0.781	0.600	0.315	0.434

Table 9. Ex-post outcome in production

This table reports the firm-level results examining the effect of receiving loans after forest loss on the firm's supply chain. In both Panel A and Panel B, the dependent variable for columns (1) and (2) is *Supply dependency*, measuring the proportion of inputs that comes from forest-dependent suppliers for each firm, calculated as the sum of sales-weighted supplier's *Dependency* for a firm. The dependent variable for columns (3) and (4) is *Country-adj supply*, measuring the proportion of inputs that comes from forest-dependent suppliers of high-deforestation-risk countries for each firm, calculated as the sum of sales-weighted supplier's *Weighted dependency* for a firm. *If get loan* is defined as one if the firm gets syndicate loan in the same year or within one year following the forest loss. Panel A uses *Post large anthropogenic loss* as the time indicator, which equals one if three years after a large human-induced forest loss, and zero if three years before. Panel B uses *Post large fire loss* as the time indicator, which equals one if three years after a large fire-induced forest loss, and zero if three years before. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A. Transition in supply chain after large human-induced loss

<i>Dependent variable</i>	<i>Supply dependency</i>		<i>Country-adj supply</i>	
	<i>+3 years</i>	<i>+4 years</i>	<i>+3 years</i>	<i>+4 years</i>
<i>Outcome window (forward)</i>	(1)	(2)	(3)	(4)
If get loan × <i>Post large anthropogenic loss</i>	-0.041* (0.020)	-0.042 (0.025)	-0.067*** (0.021)	-0.067** (0.029)
If get loan	0.052** (0.021)	0.055** (0.023)	0.094** (0.035)	0.096** (0.037)
<i>Post large anthropogenic loss</i>	0.040* (0.021)	0.036 (0.021)	0.055 (0.033)	0.051 (0.034)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	523	523	523	523
Adjusted R-squared	0.368	0.385	0.423	0.441

Panel B. Transition in supply chain after large forest loss from fire

<i>Dependent variable</i>	<i>Supply dependency</i>		<i>Country-adj supply</i>	
	<i>+3 years</i>	<i>+4 years</i>	<i>+3 years</i>	<i>+4 years</i>
<i>Outcome window (forward)</i>	(1)	(2)	(3)	(4)
If get loan × <i>Post large fire loss</i>	-0.027** (0.012)	-0.025* (0.013)	-0.037 (0.024)	-0.034 (0.024)
If get loan	0.053*** (0.018)	0.052*** (0.018)	0.074** (0.032)	0.071** (0.031)
<i>Post large fire loss</i>	0.005 (0.016)	0.004 (0.015)	0.002 (0.027)	0.001 (0.026)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	615	615	615	615
Adjusted R-squared	0.289	0.303	0.340	0.357

Table 10. Ex-post outcome in reforestation

This table reports the firm-level results examining the effect of receiving loans after human-induced loss on the reforestation around a firm. In both Panel A and Panel B, the dependent variable is *NDVI*, measuring the greenness of vegetation of the land surface around a firm on a scale from 0 to 100. The time indicator is *Post large anthropogenic loss*, which equals one if three years after a large human-induced forest loss, and zero if three years before. *If get loan* is defined as one if the firm gets syndicate loan in the same year or within one year following the forest loss. Panel A report the tests using full sample, and Panel B report the subsample of firms with forest dependency score above median. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A. Outcome in reforestation: Full sample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
If get loan × <i>Post large anthropogenic loss</i>	1.145*** (0.313)	1.184*** (0.340)	1.168*** (0.342)	1.093*** (0.347)
If get loan	-0.183 (0.660)	-0.145 (0.651)	-0.108 (0.648)	-0.051 (0.645)
<i>Post large anthropogenic loss</i>	-2.099 (1.238)	-1.879 (1.270)	-1.733 (1.292)	-1.595 (1.279)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	4,322	4,322	4,322	4,322
Adjusted R-squared	0.225	0.225	0.221	0.216

Panel B. Outcome in reforestation: High forest dependency subsample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
If get loan × <i>Post large anthropogenic loss</i>	1.724** (0.617)	1.736** (0.642)	1.751** (0.642)	1.677** (0.650)
If get loan	0.595 (1.001)	0.669 (0.993)	0.738 (0.992)	0.812 (0.998)
<i>Post large anthropogenic loss</i>	-2.117 (1.616)	-1.823 (1.638)	-1.744 (1.646)	-1.617 (1.607)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	2,297	2,297	2,297	2,297
Adjusted R-squared	0.244	0.242	0.237	0.233

Table 11. Ex-post outcome in divesting pollutive plants

This table reports the firm-level results examining the effect of receiving loans after human-induced loss on divesting pollutive plants. The dependent variable is *Divestiture*, an indicator that equals 100 if the firm divests a forest-dependent pollutive plant in the following estimation window. *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. *If get loan* is an indicator defined as one if the firm gets syndicate loan in a year, and zero otherwise. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Divestiture</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
<i>Anthropogenic loss</i> × <i>Dependency</i> × <i>If get loan</i>	0.263 (0.193)	1.107*** (0.118)	1.132*** (0.162)	1.234*** (0.263)
<i>Anthropogenic loss</i> × <i>Dependency</i>	0.285 (0.260)	0.551 (0.360)	1.039* (0.532)	1.417** (0.672)
<i>Anthropogenic loss</i> × <i>If get loan</i>	0.181 (0.110)	-0.409*** (0.143)	-0.524*** (0.149)	-0.468*** (0.150)
<i>Dependency</i> × <i>If get loan</i>	-0.636** (0.280)	-1.011*** (0.263)	-1.056*** (0.266)	-1.307*** (0.377)
<i>Anthropogenic loss</i>	-0.145** (0.060)	-0.112 (0.091)	-0.240 (0.148)	-0.440* (0.221)
<i>Dependency</i>	0.014 (0.261)	-0.325 (0.287)	-0.635 (0.406)	-0.847 (0.527)
<i>If get loan</i>	0.213 (0.242)	0.586** (0.278)	0.881*** (0.286)	0.837** (0.332)
Constant	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	7,313	7,313	7,313	7,313
Adjusted R-squared	0.004	0.014	0.021	0.026

Appendix

Table A.1. Variable definition

Variable	Definition	Source
Acute physical risk	The frequency of mentions of the unigrams or bigrams related to the acute climate discussion in the proximity of risk synonyms in the previous year earnings call transcript, divided by the total length of the transcript and standardized.	Li et al. (2024), StreetEvents
Anthropogenic loss	The size of forest loss from human activities within 10km around a firm's headquarter in the previous year in km ² .	Hansen et al. (2013), Tyukavina et al. (2022)
Bank size	The log value of total assets (million USD) of a bank in year t-1.	Compustat, Refinitiv, BankFocus
Book value of debt	The sum of long-term debt, notes payable, and the current portion of long-term debt.	Compustat, Refinitiv
Cash flow	Operating cash flow divided by previous year tangible assets of a firm in year t-1.	Compustat, Refinitiv
Chronic physical risk	The frequency of mentions of the unigrams or bigrams related to the chronic climate discussion in the proximity of risk synonyms in the previous year earnings call transcript, divided by the total length of the transcript and standardized.	Li et al. (2024), StreetEvents
Country-adj supply	The proportion of inputs that comes from forest-dependent suppliers of high-deforestation-risk countries for each firm, calculated as the sum of sales-weighted supplier's <i>Weighted dependency</i> for a firm.	Compustat Segment
Dependency	Level of production processes' dependency on forest of a firm at the 2-digit SIC level on a 0-5 scale.	ENCORE
Divestiture	An indicator that equals 100 if the firm divests a forest-dependent pollutive plant in the following estimation window.	TRI EPA, SDC M&A
Fire loss	The area of forest loss from fire within 10km around a firm's headquarter in the previous year in km ² .	Tyukavina et al. (2022)
Firm commit	The total number of firm disclosures that mentioned "deforestation" in the previous year. Corporate filings include: ESG reports, annual reports; SEC filings, etc.	Refinitiv AdvFil
Firm size	The log value of total assets (million USD) of a firm in year t-1.	Compustat, Refinitiv
High dependency	An indicator that equals one if the firm's dependent score <i>Dependency</i> is above the sample median, and zero otherwise.	ENCORE
If base prime	An indicator that equals one if the base rate for a loan is the prime rate rather than LIBOR, and zero otherwise.	DealScan
If fire	An indicator that equals one if a firm has non-zero fire-induced loss in the previous year, and zero otherwise.	Tyukavina et al. (2022)
If get loan	An indicator that equals one if the firm gets syndicate loan in the same year or within one year following the forest loss, and zero otherwise.	DealScan, Hansen et al. (2013), Tyukavina et al. (2022)
If refinance	An indicator that equals one if the loan refinances a previous loan, and zero otherwise.	DealScan
If secured loan	An indicator that equals one if the loan tranche (facility) is secured, and zero otherwise.	DealScan
Leverage	Total liabilities divided by total assets of a firm in year t-1.	Compustat, Refinitiv
Liquidity	Cash divided by total assets of a firm in year t-1.	Compustat, Refinitiv
Loan purpose	An indicator that equals one if the loan tranche (facility) purpose is M&A, and zero otherwise.	DealScan
Log amount	The log value of loan amount (million USD).	DealScan
Log credit rating	Moody's short-term issuer rating of a firm in year t-1.	Refinitiv
Log maturity	The log value of loan maturity (month).	DealScan

NDVI	MODIS normalized difference vegetation index (NDVI) within 10km around the firm's headquarter, measuring how "green" a vegetation area is from -1 to 1. The measure is normalized to 0-100 scale in the regressions.	MODIS NDVI
Post large anthropogenic loss	A time indicator that equals one if the period within estimation window is after a large human-induced forest loss, and zero if before.	Hansen et al. (2013), Tyukavina et al. (2022)
Post large fire loss	A time indicator that equals one if the period within estimation window is after a large fire-induced forest loss, and zero if before.	Tyukavina et al. (2022)
Post EUDR	A time indicator defined as one after the first deforestation framework stage (July 2019), and zero otherwise.	European Commission
Post EUDR (phase 1)	A time indicator defined as one for the period between the first deforestation policy framework and the regulation's entry into force (June 2023), and zero otherwise.	European Commission
Post EUDR (phase 2)	A time indicator defined as one for the period after the EUDR goes into force (June 2023), and zero otherwise.	European Commission
Reforest	An indicator that equals 100 if the firm claims to engage in reforestation activities in their disclosures in the following estimation window.	Refinitiv AdvFil
Repeated lending	An indicator that equals 1 if there is a past relationship with any of the lead banks in the last five years before the present loan and 0 otherwise.	DealScan
ROA	Net profit divided by total assets in year t-1.	Refinitiv, Compustat
Supply dependency	The proportion of inputs that comes from forest-dependent suppliers for each firm, calculated as the sum of sales-weighted supplier's <i>Dependency</i> for a firm.	Compustat Segment
Top dependency	An indicator that equals one if the firm's dependent score <i>Dependency</i> is above the top 30% of the sample, and zero if below bottom 30%.	ENCORE
Transition risk	The frequency of mentions of the unigrams or bigrams related to the transition climate discussion in the proximity of risk synonyms in the previous year earnings call transcript, divided by the total length of the transcript and standardized.	Li et al. (2024), StreetEvents
Weighted dependency	Country-year-level forest loss-weighted dependency score in year t-1, rescaled to a range of 0 to 5. The measure is calculated as $Dependency \times (1 + forest\ loss_{c,t})$, where $forest\ loss_{c,t}$ is the country-year total area of forest loss. Both <i>Dependency</i> and $forest\ loss_{c,t}$ are normalized to 0-1 during the calculation.	ENCORE, GLAD
Yield spread	All-in-spread drawn (AISD) divided by 100.	DealScan

Online Appendix

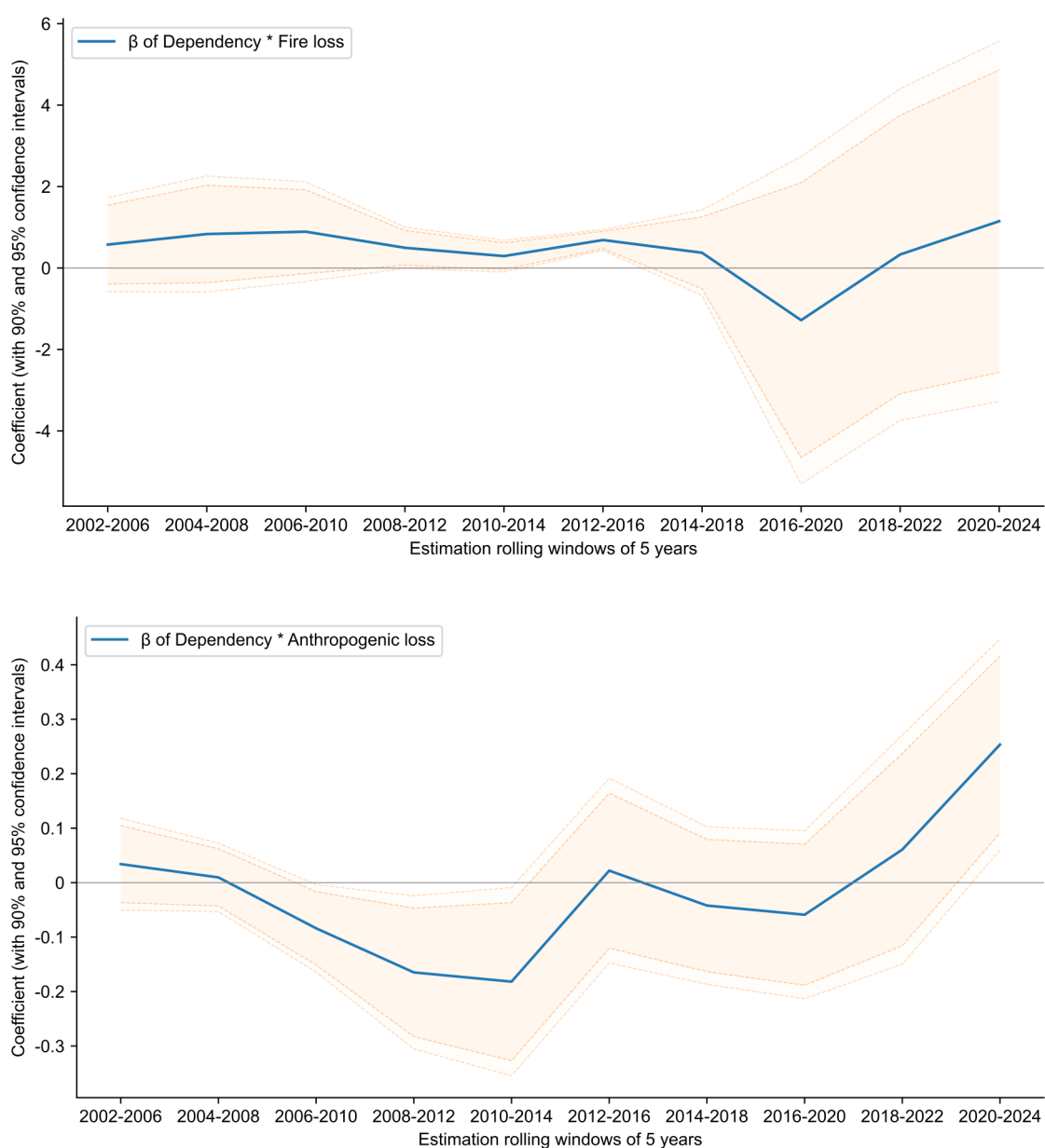


Figure B.1. Year-by-year regression: Effect of forest loss on yield spread

This figure shows estimated interaction coefficients between *Dependency* and forest loss in year-by-year loan-level regressions where *Yield spread* is the dependent variable. Each regression is run on a five-year subsample, rolling in two-year intervals from 2002-2006 to 2020-2024. The top panel plots coefficients using fire-induced forest loss, and the bottom panel uses human-induced forest loss. The blue line shows coefficient estimates. Dark orange shading indicates statistical significance at the 10% level, and light orange at the 5% level.

Table B.1 Robustness: Pricing in samples with more localized operating exposure

This table reports loan-level regression results for restricted subsamples in which firm-level exposure to forest loss is more tightly linked to borrower operations. Columns (1) and (3) restrict the sample to borrowers matched to Orbis manufacturing firms classified as single headquarters with no branches. Columns (2) and (4) restrict the sample to borrowers in the bottom decile of total assets in the full sample. The dependent variable is *Yield spread*. Columns (1) and (2) use *Fire loss* as the loss measure, which is the firm-level forest loss from fire in the previous year. Columns (3) and (4) use *Anthropogenic loss*, which is the firm-level forest loss from human activities in the previous year. All columns use *Dependency* to measure forest dependency at the 2-digit SIC level. *Post EUDR* is a time indicator, defined as one after the deforestation framework stage (July 2019), and zero otherwise. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Loss measure</i>	<i>Fire loss</i>		<i>Anthropogenic loss</i>	
<i>Subsample</i>	<i>Single HQ</i>	<i>Small firms</i>	<i>Single HQ</i>	<i>Small firms</i>
	(1)	(2)	(3)	(4)
Loss measure×Dependency	6.360** (2.457)	2.760*** (0.679)	-0.051 (0.251)	-0.224** (0.080)
Loss measure×Dependency× <i>Post EUDR</i>			1.494** (0.551)	3.227** (1.194)
Loss measure× <i>Post EUDR</i>			-0.963* (0.500)	0.515 (0.882)
Dependency× <i>Post EUDR</i>			-0.187** (0.088)	0.247 (0.389)
<i>Post EUDR</i>			-0.307* (0.160)	-1.873*** (0.575)
Loss measure	-2.645 (2.221)	-0.937 (1.116)	0.185 (0.258)	0.293*** (0.089)
Dependency	-0.007 (0.058)	-0.052 (0.119)	0.037 (0.067)	-0.056 (0.113)
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	4,766	931	4,766	931
Adjusted R-squared	0.211	0.209	0.207	0.199

Table B.2. Forest loss and loan spreads controlling for climate risk index

This table reports the robustness results controlling the climate risk index of Li et al. (2024) for the tests examining how forest loss impacts yield spreads differently for firms based on their forest dependency. The regression sample is the US listed firms with call transcripts in a year. The dependent variable is *Yield spread*, measured by AISD divided by 100. Columns (1) and (2) use *Dependency* to measure forest dependency at the 2-digit SIC level, and columns (3) to (4) use *Weighted dependency*, which adjusts for risks associated with country-year-level forest loss and is rescaled to a range of 0 to 5. *Fire loss* refers to firm-level forest loss from fire in the previous year, and *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. *Acute physical risk*, *Chronic physical risk*, and *Transition risk* are three climate risk proxies calculated by the frequency of mentions of the unigrams or bigrams related to the acute, chronic, and transition climate discussion, respectively, in the previous year earnings call transcript, divided by the total length of the transcript and standardized (Li et al. 2024). All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. Definitions for all variables are provided in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC level and year level, with values reported in parentheses. Statistical significance is denoted by ***, **, and * at the 1%, 5%, and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Dependency measures</i>	<i>Dependency</i>		<i>Weighted dependency</i>	
	(1)	(2)	(3)	(4)
Fire loss×Dependency measure	0.653*** (0.132)	0.651*** (0.135)	0.862*** (0.173)	0.860*** (0.176)
Anthropogenic loss×Dependency measure	-0.081 (0.070)	-0.083 (0.070)	-0.065 (0.081)	-0.067 (0.081)
Fire loss	-0.310 (0.246)	-0.328 (0.248)	-0.362 (0.261)	-0.381 (0.263)
Anthropogenic loss	0.135 (0.082)	0.135 (0.080)	0.119 (0.087)	0.118 (0.084)
Dependency measure	-0.008 (0.076)	-0.010 (0.076)	-0.031 (0.082)	-0.034 (0.082)
<i>Acute physical risk</i>		0.014 (0.061)		0.015 (0.062)
<i>Chronic physical risk</i>		0.008 (0.033)		0.008 (0.032)
<i>Transition risk</i>		-0.072*** (0.022)		-0.073*** (0.022)
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	4,277	4,277	4,277	4,277
Adjusted R-squared	0.374	0.375	0.374	0.375

Table B.3. Forest loss and loan spreads: Fire cut-offs

This table reports the loan-level results with the interaction between the equally distributed forest dependency groups and fire-induced loss measures, to further examine the loan pricing effects between high versus low forest-dependency groups following fire-induced forest loss. The dependent variable is *Yield spread*, measured by AISD divided by 100. Forest dependency group is separated by *High dependency*, defined as one if the firm's dependent score is above the sample median and zero otherwise. Column (1) uses *Fire loss* as the loss measure, which is the firm-level forest loss from fire in the previous year in km². Column (2) uses *If fire*, which equals one if a firm has non-zero fire-induced loss in the previous year, and zero otherwise. Columns (3) to (6) defines the top fire loss indicators based on percentile cut-offs of top 40%, 25%, 5%, and 3%, respectively, to compare with situations of no fire loss. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>						
<i>Fire measure</i>	<i>Fire loss</i>	<i>If fire</i>	<i>Top 40%</i>	<i>Top 25%</i>	<i>Top 5%</i>	<i>Top 3%</i>
	(1)	(2)	(3)	(4)	(5)	(6)
Fire measure×High dependency	0.190** (0.086)	0.146** (0.065)	0.158** (0.076)	0.189* (0.092)	0.237** (0.098)	0.376** (0.143)
Fire measure	-0.052 (0.045)	-0.043 (0.058)	-0.033 (0.069)	-0.032 (0.074)	0.055 (0.039)	-0.054 (0.055)
High dependency	-0.208** (0.078)	-0.284*** (0.101)	-0.280** (0.106)	-0.283** (0.102)	-0.248** (0.106)	-0.234** (0.105)
<i>Statistics: fire loss cut-off group</i>						
Mean (km ²)		0.01113	0.02763	0.04315	0.18511	0.28869
Equivalent # of football pitches		2.08	5.16	8.07	34.60	53.96
Standard deviation		0.1955	0.3084	0.3893	0.8558	1.0917
# of firm-year loss cases		23,411	17,977	10,922	2,004	1,221
Constant	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	42,590	42,590	37,136	30,065	21,125	20,342
Adjusted R-squared	0.359	0.360	0.357	0.358	0.352	0.348

Table B.4. Country heterogeneity of EUDR's effect: EU operators

This table presents the robustness results on how the EUDR introduction affects yield spreads for forest-dependent firms compared to other firms when human-induced loss occurred. The analysis examines how EU banks and non-EU banks respond differently to EU operators. The dependent variable is *Yield spread*. Columns (1) and (2) use *Dependency* to measure forest dependency, and columns (3) and (4) use *Weighted dependency* to account for country-year-level forest loss risk. *Post EUDR* is a time indicator, defined as one after the deforestation framework stage (July 2019), and zero otherwise. Columns (1) and (3) examine EU banks with EU operators, and Columns (2) and (4) examine non-EU banks with EU operators. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Dependency measure</i>	<i>Dependency</i>		<i>Weighted dependency</i>	
<i>Bank country</i>	<i>EU bank</i>	<i>Non-EU</i>	<i>EU bank</i>	<i>Non-EU</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss × Dependency measure × <i>Post EUDR</i>	1.477** (0.566)	-0.422 (1.574)	2.162 (1.411)	-0.936 (2.663)
Anthropogenic loss × Dependency measure	0.031 (0.176)	0.062 (0.229)	0.026 (0.288)	0.079 (0.390)
Anthropogenic loss × <i>Post EUDR</i>	-1.560** (0.741)	0.621 (1.333)	-1.315 (0.956)	0.767 (1.384)
Dependency measure × <i>Post EUDR</i>	0.103 (0.110)	0.649* (0.344)	0.112 (0.279)	0.954 (0.574)
Anthropogenic loss	0.074 (0.181)	-0.062 (0.212)	0.093 (0.181)	-0.058 (0.235)
Dependency measure	-0.225*** (0.072)	-0.035 (0.074)	-0.276* (0.158)	0.057 (0.145)
<i>Post EUDR</i>	0.324* (0.159)	-0.971*** (0.342)	0.356* (0.184)	-0.904** (0.351)
<i>Chi-square test</i>	(1) = (2)	4.645**	(3) = (4)	3.819*
<i>P-value</i>		0.031		0.051
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	6,171	5,671	6,171	5,671
Adjusted R-squared	0.479	0.450	0.475	0.449

Table B.5. EU deforestation regulation: Phase 1 and 2

This table reports the results examining the effects of different phases of EUDR on yield spread for forest-dependent firms and other firms when human-induced loss occurred. The dependent variable is *Yield spread*. Columns (1) and (2) use *Dependency* to measure forest dependency, and columns (3) and (4) use *Weighted dependency* to account for country-year-level forest loss risk. *Post EUDR (phase 1)* is the time indicator defined as one for the period between the deforestation policy framework and the regulation's entry into force (June 2023), and zero otherwise. *Post EUDR (phase 2)* is the time indicator defined as one after the EUDR goes into force, and zero otherwise. Columns (1) and (3) examine pairs of EU banks with EU firms, and Columns (2) and (4) examine other lending situations. Loan controls include *If secured loan*, *If base prime*, *If refinance*, *Repeated lending*, *Loan purpose*, *Log loan amount*, and *Log maturity*. Borrower controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Lender controls include *Bank size*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Yield spread</i>				
<i>Dependency measure</i>	<i>Dependency</i>		<i>Weighted dependency</i>	
<i>Bank country – Firm country pair</i>	<i>EU</i>	<i>Non-EU</i>	<i>EU</i>	<i>Non-EU</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency measure	1.972***	0.229	3.662***	0.324
<i>Post EUDR (phase 1)</i>	(0.616)	(0.191)	(1.160)	(0.241)
Anthropogenic loss×Dependency measure	4.000**	0.167	7.515**	0.274
<i>Post EUDR (phase 2)</i>	(1.530)	(0.109)	(2.961)	(0.262)
Anthropogenic loss×Dependency measure	-0.034	-0.032	-0.068	-0.073**
	(0.171)	(0.031)	(0.319)	(0.029)
Anthropogenic loss× <i>Post EUDR (phase 1)</i>	-1.642**	-0.333	-1.617**	-0.324
	(0.645)	(0.240)	(0.641)	(0.205)
Anthropogenic loss× <i>Post EUDR (phase 2)</i>	-9.630***	-0.276	-9.582***	-0.258
	(1.226)	(0.270)	(1.262)	(0.308)
Dependency measure× <i>Post EUDR (phase 1)</i>	-0.084	0.174	-0.136	0.212
	(0.291)	(0.119)	(0.563)	(0.191)
Dependency measure× <i>Post EUDR (phase 2)</i>	-0.179	0.050	-0.346*	0.090
	(0.104)	(0.122)	(0.198)	(0.179)
Anthropogenic loss	0.128	0.064*	0.131	0.085**
	(0.187)	(0.034)	(0.187)	(0.038)
Dependency measure	-0.237***	0.023	-0.447***	0.151**
	(0.074)	(0.047)	(0.139)	(0.070)
<i>Post EUDR (phase 1)</i>	0.593**	0.029	0.583**	0.073
	(0.215)	(0.155)	(0.220)	(0.150)
<i>Post EUDR (phase 2)</i>	1.788***	0.511***	1.781***	0.503***
	(0.206)	(0.144)	(0.206)	(0.129)
<i>Chi-square test (phase 1)</i>	(1) = (2)	5.720**	(3) = (4)	5.938**
<i>Chi-square test (phase 2)</i>	(1) = (2)	14.330***	(3) = (4)	14.243***
Constant	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Loan Controls	Yes	Yes	Yes	Yes
Borrower Controls	Yes	Yes	Yes	Yes
Lender Controls	Yes	Yes	Yes	Yes
Observations	5,518	37,072	5,518	37,072
Adjusted R-squared	0.491	0.358	0.491	0.359

Table B.6. Ex-post outcome in reforestation after fire-induced loss

This table reports the firm-level results examining the effect of receiving loans after fire-induced loss on the reforestation around a firm. In both Panel A and Panel B, the dependent variable is *NDVI*, measuring the greenness of vegetation of the land surface around a firm on a scale from 0 to 100. The time indicator is *Post large fire loss*, which equals one if three years after a large fire-induced forest loss, and zero if three years before. *If get loan* is defined as one if the firm gets syndicate loan in the same year or within one year following the forest loss. Panel A report the tests using full sample, and Panel B report the subsample of firms with forest dependency score above median. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A. Outcome in reforestation: Full sample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
If get loan × <i>Post large fire loss</i>	-0.168 (0.523)	-0.103 (0.512)	-0.064 (0.519)	-0.042 (0.536)
If get loan	-0.003 (0.816)	-0.053 (0.819)	-0.087 (0.817)	-0.120 (0.815)
<i>Post large fire loss</i>	0.051 (2.411)	0.537 (2.467)	0.839 (2.502)	0.876 (2.506)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	6,811	6,811	6,811	6,811
Adjusted R-squared	0.073	0.073	0.072	0.072

Panel B. Outcome in reforestation: High forest dependency subsample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
If get loan × <i>Post large fire loss</i>	-0.382 (0.554)	-0.315 (0.534)	-0.284 (0.542)	-0.285 (0.565)
If get loan	0.709 (0.978)	0.652 (0.986)	0.623 (0.984)	0.603 (0.983)
<i>Post large fire loss</i>	-0.885 (2.353)	-0.466 (2.402)	-0.194 (2.420)	-0.168 (2.427)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	3,642	3,642	3,642	3,642
Adjusted R-squared	0.095	0.095	0.095	0.095

Table B.7. Loan receipt and differences in firms' book value of debt

This table compares the book value of debt between firms that receive bank syndicate loans in a year (*If get loan* = 1) and those that do not (*If get loan* = 0). The book value of debt is measured as the sum of long-term debt, notes payable, and the current portion of long-term debt.

	If get loan=1		If get loan=0		Diff: Yes - No
	Mean (std. dev.)	Obs	Mean (std. dev.)	Obs	Mean (std. err.)
Book value of debt (m\$)	33,126.102 (94,882.800)	23,891	28,191.304 (86,935.547)	93,292	4,934.7978*** (642.5301)

Table B.8. Selection into granting loans

This table reports the first-stage probit model estimating the likelihood of a firm receiving a syndicated loan in a given year. The dependent variable *If get loan* is an indicator equal to one if the firm obtains a syndicated loan and zero otherwise. Firm-level controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. Columns (2) and (3) include *Anthropogenic loss* and *Dependency*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: If get loan</i>			
	(1)	(2)	(3)
Firm size	0.027*** (0.004)	0.028*** (0.004)	0.028*** (0.004)
Leverage	0.178*** (0.032)	0.183*** (0.034)	0.183*** (0.034)
ROA	0.238*** (0.035)	0.262*** (0.037)	0.261*** (0.037)
Liquidity	-0.518*** (0.051)	-0.503*** (0.054)	-0.500*** (0.056)
Log credit rating	0.066*** (0.007)	0.064*** (0.008)	0.064*** (0.008)
Anthropogenic loss		0.025* (0.013)	0.024* (0.013)
Dependency			0.018** (0.008)
Constant	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Observations	104,166	99,198	99,198
Adjusted R-squared	0.030	0.031	0.031

Table B.9. Robustness: Ex-post outcome in production

This table reports the robustness results examining the effect of receiving loans after forest loss on the firm's supply chain, controlling for selection bias using the Inverse Mills Ratio (IMR) from the first-stage probit model (Table B.8). In both Panel A and Panel B, the dependent variable in columns (1) and (2) is *Supply dependency*, measuring the proportion of inputs sourced from forest-dependent suppliers. Columns (3) and (4) use *Country-adj supply*, capturing inputs from forest-dependent suppliers in high-deforestation-risk countries. *If get loan* equals one if the firm secures a syndicated loan in the same year or within one year after forest loss. Panel A defines the post-event period based on large human-induced forest loss, while Panel B uses large fire-induced forest loss. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, *Log credit rating*, and the IMR term. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Transition in supply chain after large human-induced loss

<i>Dependent variable</i>	<i>Supply dependency</i>		<i>Country-adj supply</i>	
	+3 years	+4 years	+3 years	+4 years
<i>Outcome window (forward)</i>	(1)	(2)	(3)	(4)
<i>If get loan</i> × <i>Post large anthropogenic loss</i>	-0.043** (0.019)	-0.044* (0.024)	-0.070*** (0.021)	-0.071** (0.028)
<i>If get loan</i>	0.052** (0.022)	0.056** (0.024)	0.096** (0.035)	0.099** (0.037)
<i>Post large anthropogenic loss</i>	0.027 (0.024)	0.025 (0.024)	0.042 (0.035)	0.040 (0.035)
IMR (<i>If get loan</i>)	-3.570 (2.588)	-3.115 (2.591)	-3.728 (4.050)	-3.238 (4.030)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	518	518	518	518
Adjusted R-squared	0.379	0.394	0.428	0.445

Panel B. Transition in supply chain after large forest loss from fire

<i>Dependent variable</i>	<i>Supply dependency</i>		<i>Country-adj supply</i>	
	+3 years	+4 years	+3 years	+4 years
<i>Outcome window (forward)</i>	(1)	(2)	(3)	(4)
<i>If get loan</i> × <i>Post large fire loss</i>	-0.027** (0.012)	-0.025* (0.013)	-0.037 (0.024)	-0.034 (0.025)
<i>If get loan</i>	0.053*** (0.018)	0.052*** (0.018)	0.074** (0.031)	0.071** (0.031)
<i>Post large fire loss</i>	0.005 (0.016)	0.004 (0.015)	0.002 (0.027)	0.001 (0.026)
IMR (<i>If get loan</i>)	0.053*** (0.018)	0.052*** (0.018)	0.074** (0.031)	0.071** (0.031)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE		Yes	Yes	Yes
Observations	614	614	614	614
Adjusted R-squared	0.289	0.303	0.340	0.357

Table B.10. Robustness: Ex-post outcome in reforestation

This table reports the robustness results examining the effect of receiving loans after human-induced forest loss on reforestation, controlling for selection bias using the Inverse Mills Ratio (IMR) from the first-stage probit model (Table B.8). In both Panel A and Panel B, the dependent variable is *NDVI*, which measures the greenness of vegetation around a firm on a scale from 0 to 100. *If get loan* equals one if the firm obtains a syndicated loan in the same year or within one year after forest loss. The time indicator is *Post large anthropogenic loss*, which equals one if three years after a large human-induced forest loss, and zero if three years before. Panel A report the tests using full sample, and Panel B report the subsample of firms with forest dependency score above median. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, *Log credit rating*, and the IMR term. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

Panel A. Outcome in reforestation: Full sample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
<i>If get loan</i> × <i>Post large anthropogenic loss</i>	0.992** (0.366)	1.047** (0.391)	1.027** (0.389)	0.961** (0.380)
<i>If get loan</i>	-0.083 (0.655)	-0.061 (0.649)	-0.024 (0.646)	0.024 (0.644)
<i>Post large anthropogenic loss</i>	-2.644** (1.241)	-2.427* (1.269)	-2.313* (1.287)	-2.181 (1.275)
IMR (<i>If get loan</i>)	-247.688*** (75.048)	-246.440*** (72.370)	-260.352*** (73.728)	-261.935*** (75.118)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	4,288	4,288	4,288	4,288
Adjusted R-squared	0.235	0.235	0.231	0.226

Panel B. Outcome in reforestation: High forest dependency subsample

<i>Dependent variable: NDVI</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
<i>If get loan</i> × <i>Post large anthropogenic loss</i>	1.686** (0.709)	1.709** (0.716)	1.717** (0.703)	1.643** (0.702)
<i>If get loan</i>	0.630 (1.086)	0.694 (1.077)	0.769 (1.073)	0.844 (1.079)
<i>Post large anthropogenic loss</i>	-2.145 (1.632)	-1.847 (1.649)	-1.772 (1.651)	-1.647 (1.612)
IMR (<i>If get loan</i>)	-16.688 (93.448)	-13.543 (92.302)	-16.040 (92.927)	-16.881 (93.407)
Constant	Yes	Yes	Yes	Yes
Firm controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	2,280	2,280	2,280	2,280
Adjusted R-squared	0.244	0.242	0.238	0.234

Table B.11. Robustness: Ex-post outcome in divesting pollutive plants

This table reports the firm-level results examining the effect of receiving loans after human-induced forest loss on the divestiture of pollutive, forest-dependent plants, controlling for selection bias using the Inverse Mills Ratio (IMR) from the first-stage probit model (Table B.8). The dependent variable is *Divestiture*, an indicator that equals 100 if the firm divests a forest-dependent pollutive plant in the following estimation window. *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. *If get loan* is an indicator defined as one if the firm gets syndicate loan in a year, and zero otherwise. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, *Log credit rating*, and the IMR term. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Divestiture</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency× <i>If get loan</i>	0.252 (0.207)	1.115*** (0.293)	1.146*** (0.305)	1.252*** (0.387)
Anthropogenic loss×Dependency	0.288 (0.272)	0.543 (0.381)	1.025* (0.574)	1.398* (0.737)
Anthropogenic loss× <i>If get loan</i>	0.192 (0.172)	-0.408** (0.195)	-0.523** (0.190)	-0.466** (0.214)
Dependency× <i>If get loan</i>	-0.646** (0.255)	-1.016** (0.409)	-1.066** (0.439)	-1.321** (0.506)
Anthropogenic loss	-0.386 (0.372)	-0.647 (0.561)	-1.191 (0.834)	-1.750 (1.101)
Dependency	-0.264 (0.439)	-0.979 (0.726)	-1.799 (1.071)	-2.450 (1.426)
<i>If get loan</i>	0.221 (0.299)	0.588 (0.355)	0.884** (0.392)	0.843* (0.476)
IMR (<i>If get loan</i>)	-28.462 (50.240)	-63.921 (61.874)	-113.649 (93.585)	-156.606 (120.255)
Constant	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	7,313	7,313	7,313	7,313
Adjusted R-squared	0.004	0.014	0.021	0.026

Table B.12. Reforestation: Alternative measure using firm disclosures

This table reports the robustness results on the effect of receiving loans after human-induced loss on firms' reforestation engagement. The dependent variable is *Reforest*, an indicator that equals 100 if the firm claims to engage in reforestation activities in their disclosures in the following estimation window. *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. *If get loan* is an indicator defined as one if the firm gets syndicate loan in a year, and zero otherwise. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, and *Log credit rating*. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Reforest</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss×Dependency× <i>If get loan</i>	0.634*** (0.115)	0.734*** (0.146)	0.684*** (0.130)	0.540*** (0.174)
Anthropogenic loss×Dependency	-0.513** (0.182)	-0.555** (0.200)	-0.444* (0.220)	-0.336 (0.270)
Anthropogenic loss × <i>If get loan</i>	0.004 (0.118)	0.032 (0.204)	0.016 (0.164)	0.065 (0.236)
Dependency × <i>If get loan</i>	-0.831*** (0.262)	-0.881** (0.363)	-1.011*** (0.341)	-1.056** (0.405)
Anthropogenic loss	0.431 (0.518)	0.589 (0.643)	0.635 (0.737)	0.672 (0.793)
Dependency	0.295 (0.297)	0.222 (0.334)	0.089 (0.351)	-0.019 (0.400)
<i>If get loan</i>	1.017*** (0.303)	1.254*** (0.340)	1.499*** (0.339)	1.560*** (0.365)
Constant	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	95,595	95,595	95,595	95,595
Adjusted R-squared	0.035	0.040	0.042	0.043

Table B.13. Robustness: Selection-adjusted results for alternative reforestation measure

This table reports the robustness results examining the effect of receiving loans after human-induced forest loss on firms' reforestation engagement, controlling for selection bias using the Inverse Mills Ratio (IMR) from the first-stage probit model (Table B.8). The dependent variable is *Reforest*, an indicator that equals 100 if the firm claims to engage in reforestation activities in their disclosures in the following estimation window. *Anthropogenic loss* refers to firm-level forest loss from human activities in the previous year. *If get loan* is an indicator defined as one if the firm gets syndicate loan in a year, and zero otherwise. Firm controls include *Firm size*, *Leverage*, *ROA*, *Liquidity*, *Log credit rating*, and the IMR term. All regressions include year fixed effects and industry fixed effects at the high-level SIC division level. All variables are defined in Appendix Table A.1. Standard errors are clustered at the 2-digit SIC and year level and reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5% and 10% levels, respectively.

<i>Dependent variable: Reforest</i>				
<i>Outcome window (forward)</i>	<i>+1 year</i>	<i>+2 years</i>	<i>+3 years</i>	<i>+4 years</i>
	(1)	(2)	(3)	(4)
Anthropogenic loss × Dependency × <i>If get loan</i>	0.632*** (0.115)	0.731*** (0.146)	0.681*** (0.141)	0.537*** (0.180)
Anthropogenic loss × Dependency	-0.517*** (0.180)	-0.560*** (0.196)	-0.450** (0.214)	-0.342 (0.263)
Anthropogenic loss × <i>If get loan</i>	-0.008 (0.119)	0.017 (0.201)	-0.001 (0.163)	0.048 (0.234)
Dependency × <i>If get loan</i>	-0.861*** (0.265)	-0.918** (0.370)	-1.051*** (0.351)	-1.100** (0.413)
Anthropogenic loss	2.722** (0.999)	3.435*** (1.182)	3.734** (1.324)	3.978*** (1.398)
Dependency	3.084*** (0.794)	3.688*** (0.964)	3.864*** (1.070)	4.007*** (1.153)
<i>If get loan</i>	1.015*** (0.306)	1.252*** (0.343)	1.496*** (0.345)	1.556*** (0.372)
IMR (<i>If get loan</i>)	269.212*** (77.910)	334.491*** (92.946)	364.226*** (101.687)	388.527*** (107.063)
Constant	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	95,595	95,595	95,595	95,595
Adjusted R-squared	0.036	0.042	0.044	0.044

References

Li Q, Shan H, Tang Y, Yao V (2024) Corporate climate risk: Measurements and responses. *The Review of Financial Studies* 37(6):1778–1830.